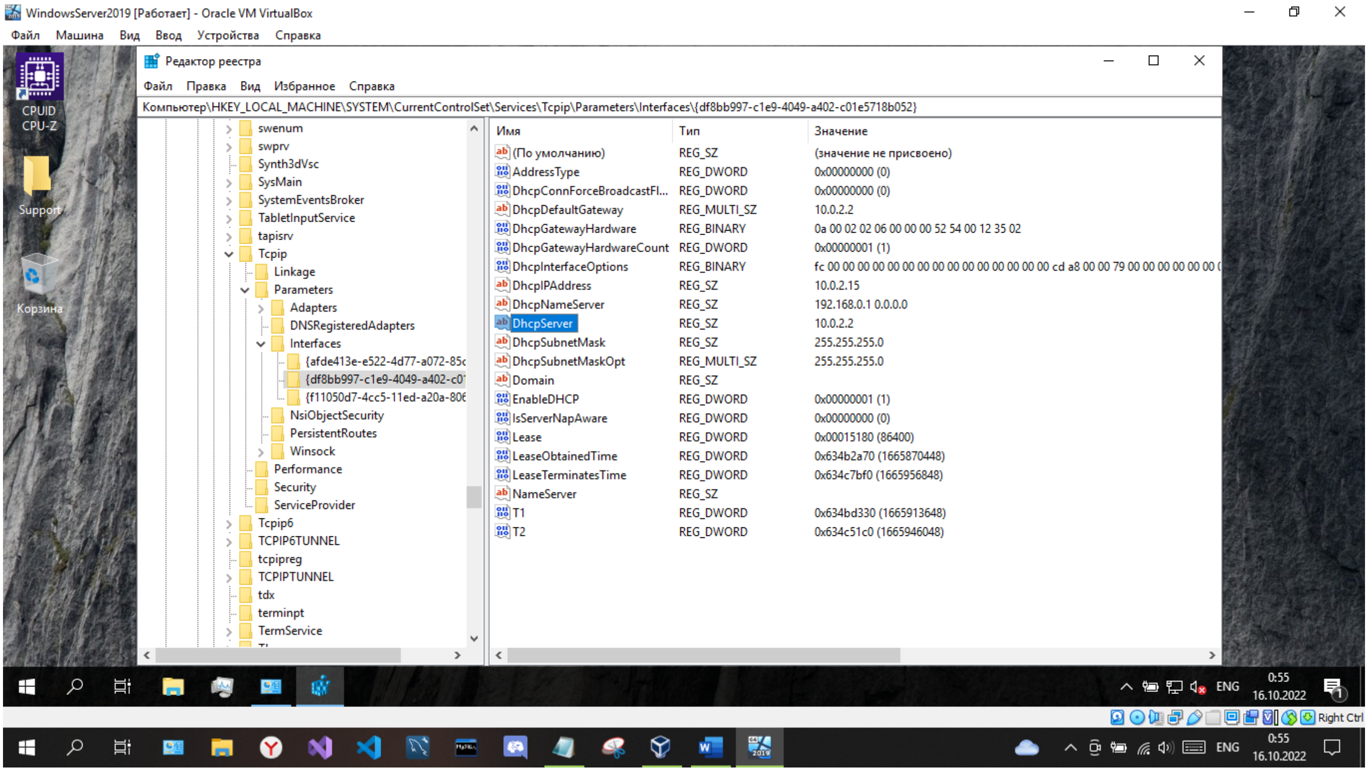
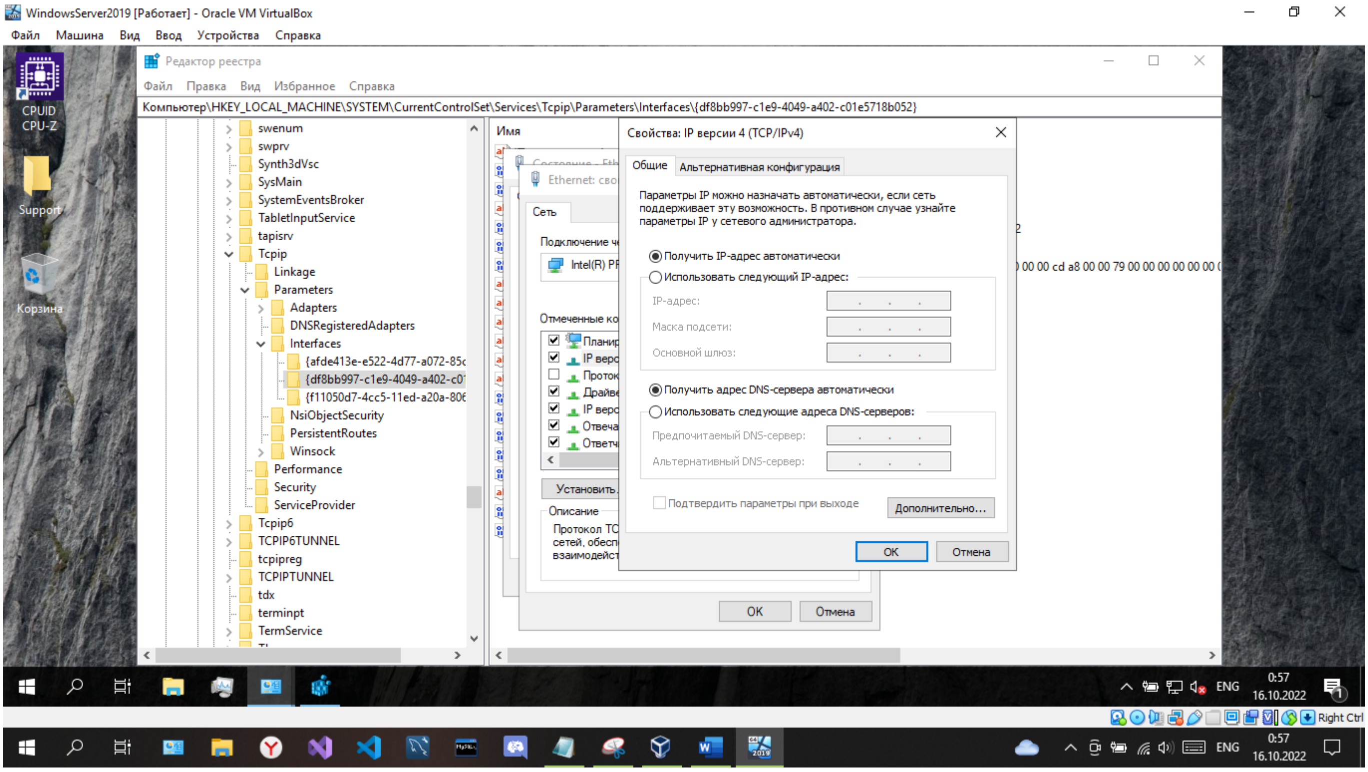
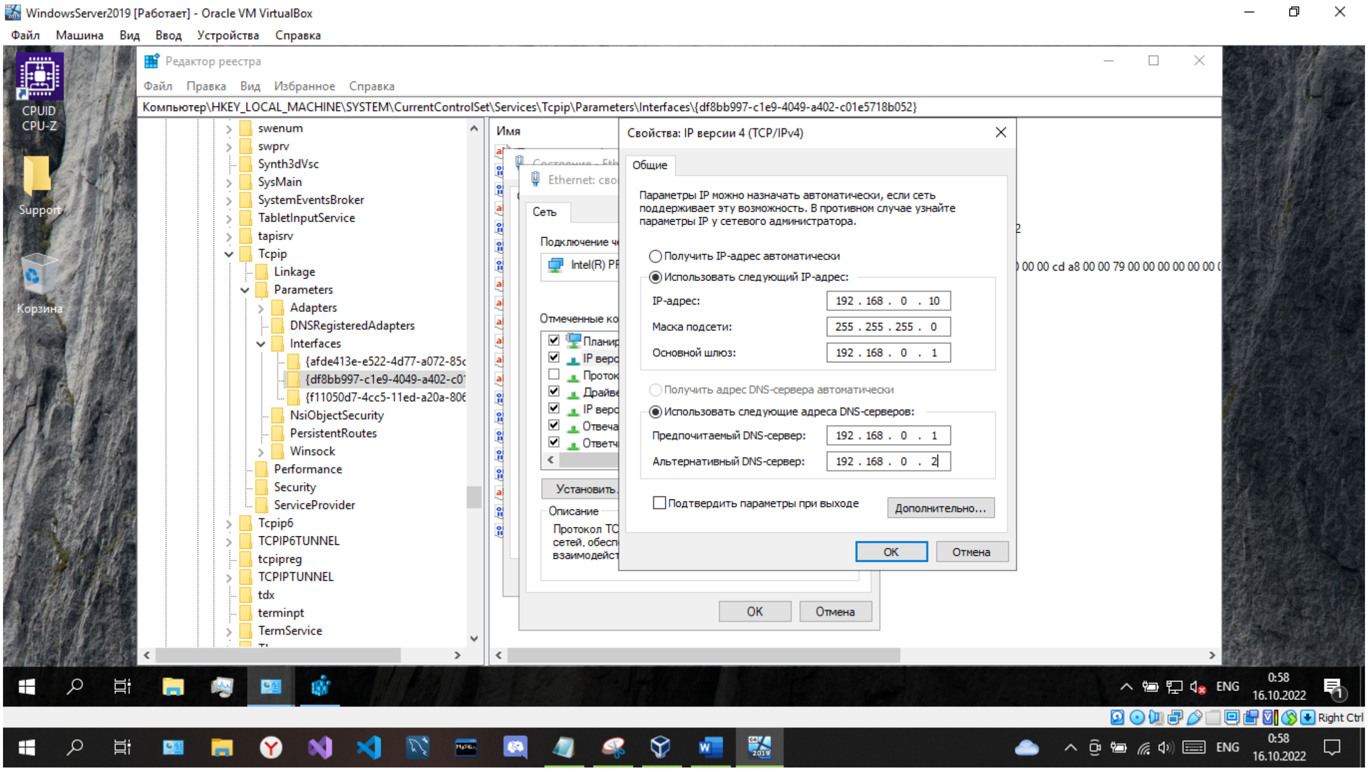
Цель работы: научиться задавать настройки TCP/IP в реестре Windows.

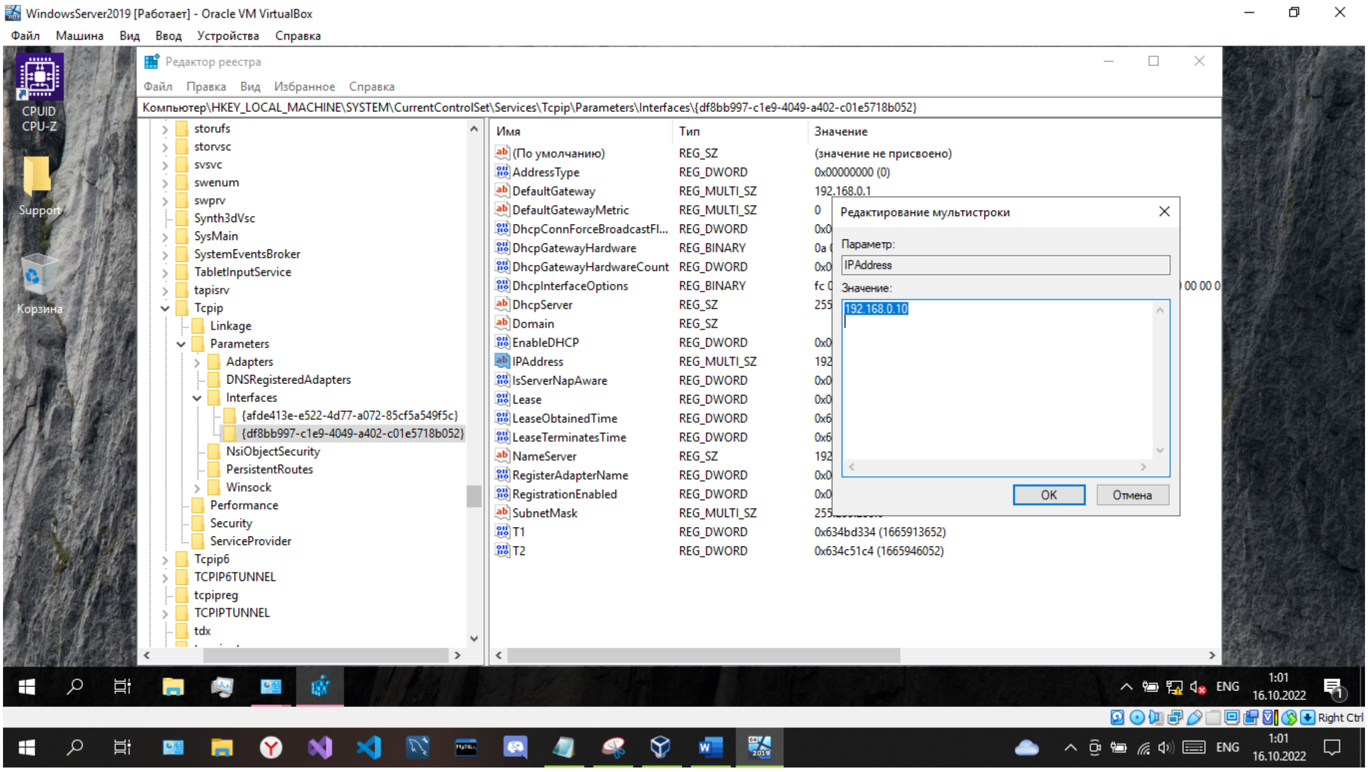
Выполнение заданий:
Вариант 11

Работа с реестром

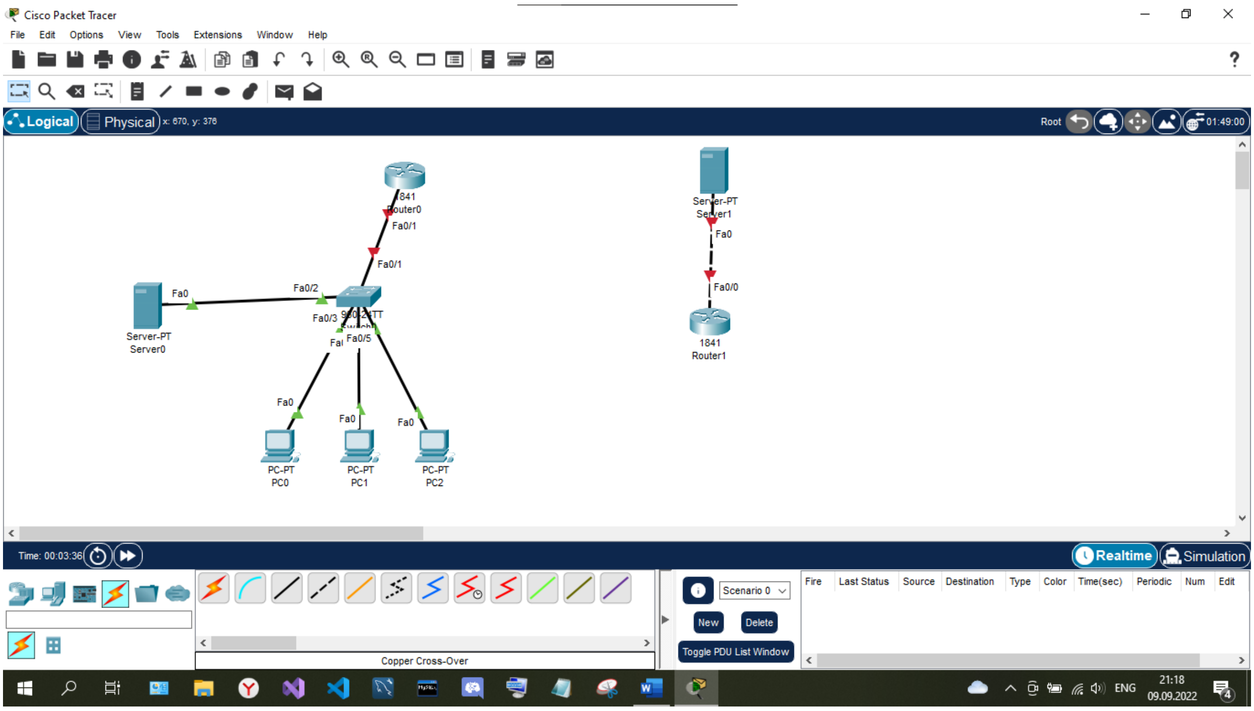





WindowsServer2019 [Работает] - Oracle VM VirtualBox
Файл Машина Вид Ввод Устройства Справка
Редактор реестра
Файл Правка Вид Избранное Справка
Компьютер\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\{df8bb997-c1e9-4049-a402-c01e5718b052}
swenum
swprv
Synth3dVsc
SysMain
SystemEventsBroker
TabletInputService
tapisrv
Tcpip
Linkage
Parameters
Adapters
DNSRegisteredAdapters
Interfaces
NsiObjectSecurity
PersistentRoutes
Winsock
Performance
Security
ServiceProvider
Tcpip6
TCPIP6TUNNEL
tcpipreg
TCPIPTUNNEL
tdx
terminpt
TermService
Свойства: IP версии 4 (TCP/IPv4)
Общие
Параметры IP можно назначать автоматически, если сеть поддерживает эту возможность. В противном случае узнайте параметры IP у сетевого администратора.
Получить IP-адрес автоматически
Использовать следующий IP-адрес:
IP-адрес: 192 . 168 . 0 . 10
Маска подсети: 255 . 255 . 255 . 0
Основной шлюз: 192 . 168 . 0 . 1
Получить адрес DNS-сервера автоматически
Использовать следующие адреса DNS-серверов:
Предпочитаемый DNS-сервер: 192 . 168 . 0 . 1
Альтернативный DNS-сервер: 192 . 168 . 0 . 2
Подтвердить параметры при выходе
Дополнительно...
OK
Отмена
ENG
0:58
16.10.2022
Right Ctrl


WindowsServer2019 [Работает] - Oracle VM VirtualBox
Редактор реестра
Компьютер\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\Interfaces\{df8bb997-c1e9-4049-a402-c01e5718b052}
Имя
Тип
Значение
(По умолчанию) REG_SZ (значение не присвоено)
AddressType REG_DWORD 0x00000000 (0)
DefaultGateway REG_MULTI_SZ 192.168.0.1
DefaultGatewayMetric REG_MULTI_SZ
DhcpConnForceBroadcastFl... REG_DWORD
DhcpGatewayHardware REG_BINARY
DhcpGatewayHardwareCount REG_DWORD
DhcpInterfaceOptions REG_BINARY
DhcpServer REG_SZ
Domain REG_SZ
EnableDHCP REG_DWORD
IPAddress REG_MULTI_SZ
IsServerNapAware REG_DWORD
Lease REG_DWORD
LeaseObtainedTime REG_DWORD
LeaseTerminatesTime REG_DWORD
NameServer REG_SZ
RegisterAdapterName REG_DWORD
RegistrationEnabled REG_DWORD
SubnetMask REG_MULTI_SZ
T1 REG_DWORD 0x634bd334 (1665913652)
T2 REG_DWORD 0x634c51c4 (1665946052)
Редактирование мультистроки
Параметр:
IPAddress
Значение:
192.168.0.10
OK
Отмена
1:01
16.10.2022

NAT



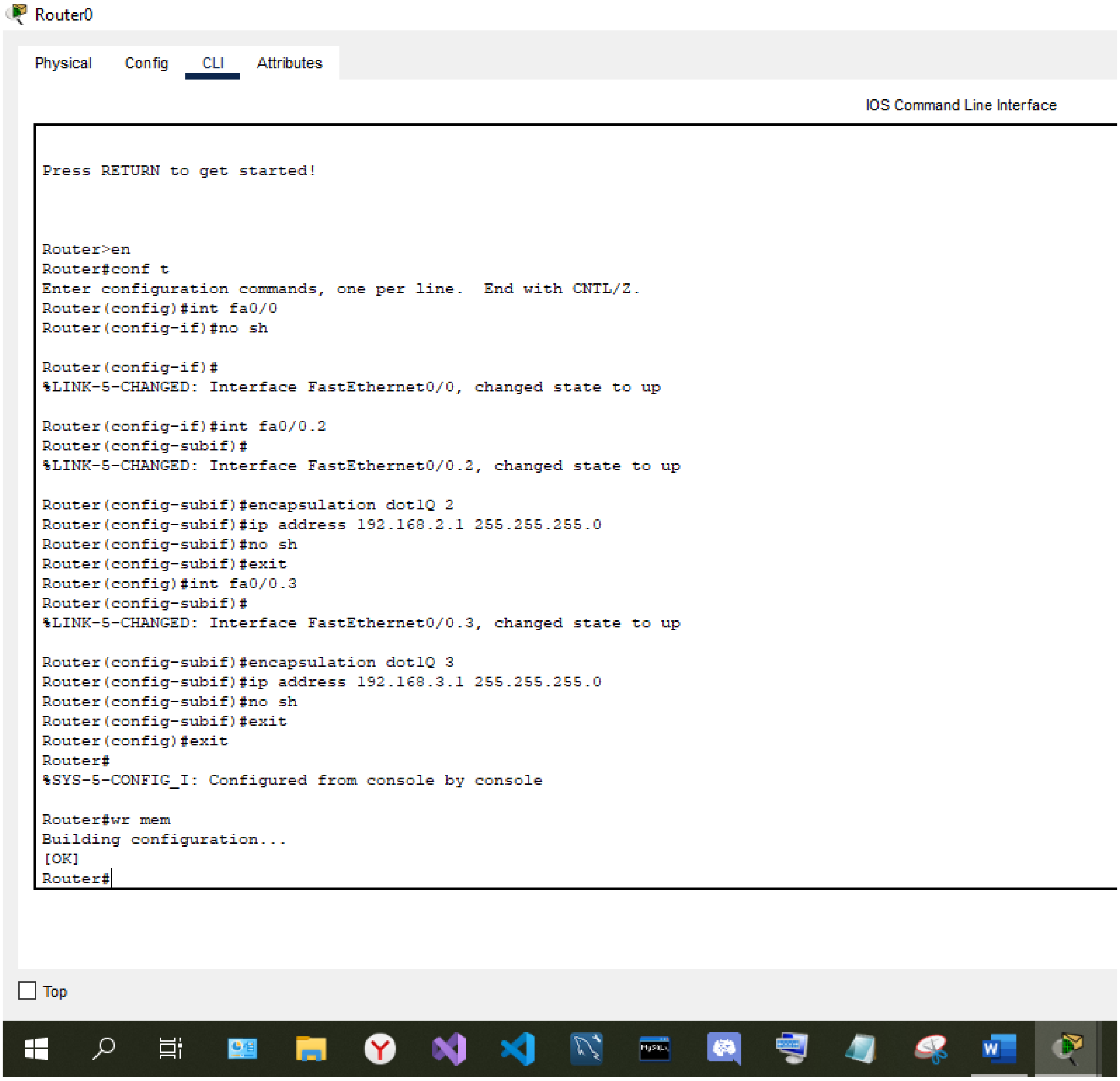
Switch0

Physical Config CLI Attributes

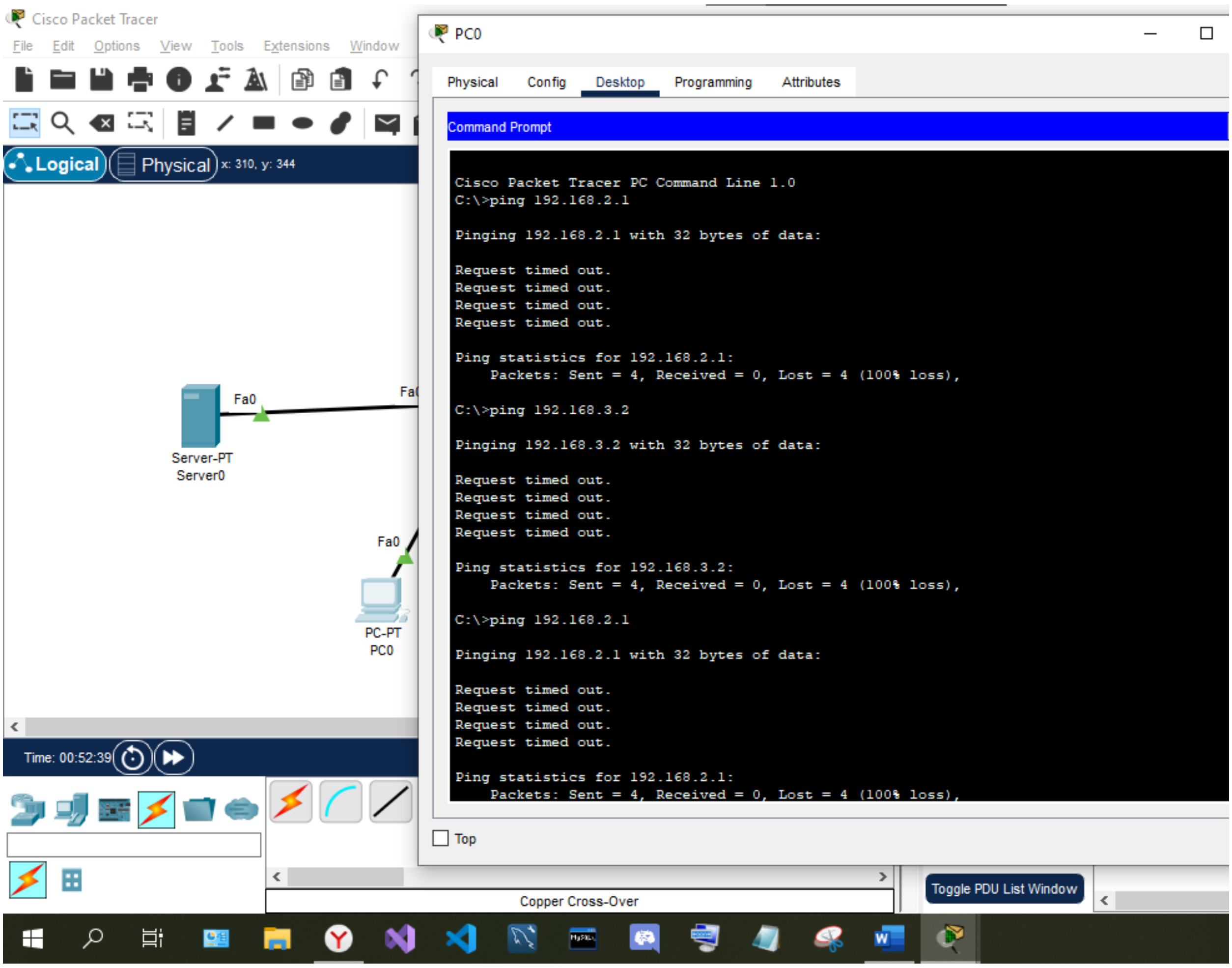
IOS Command Line Interface

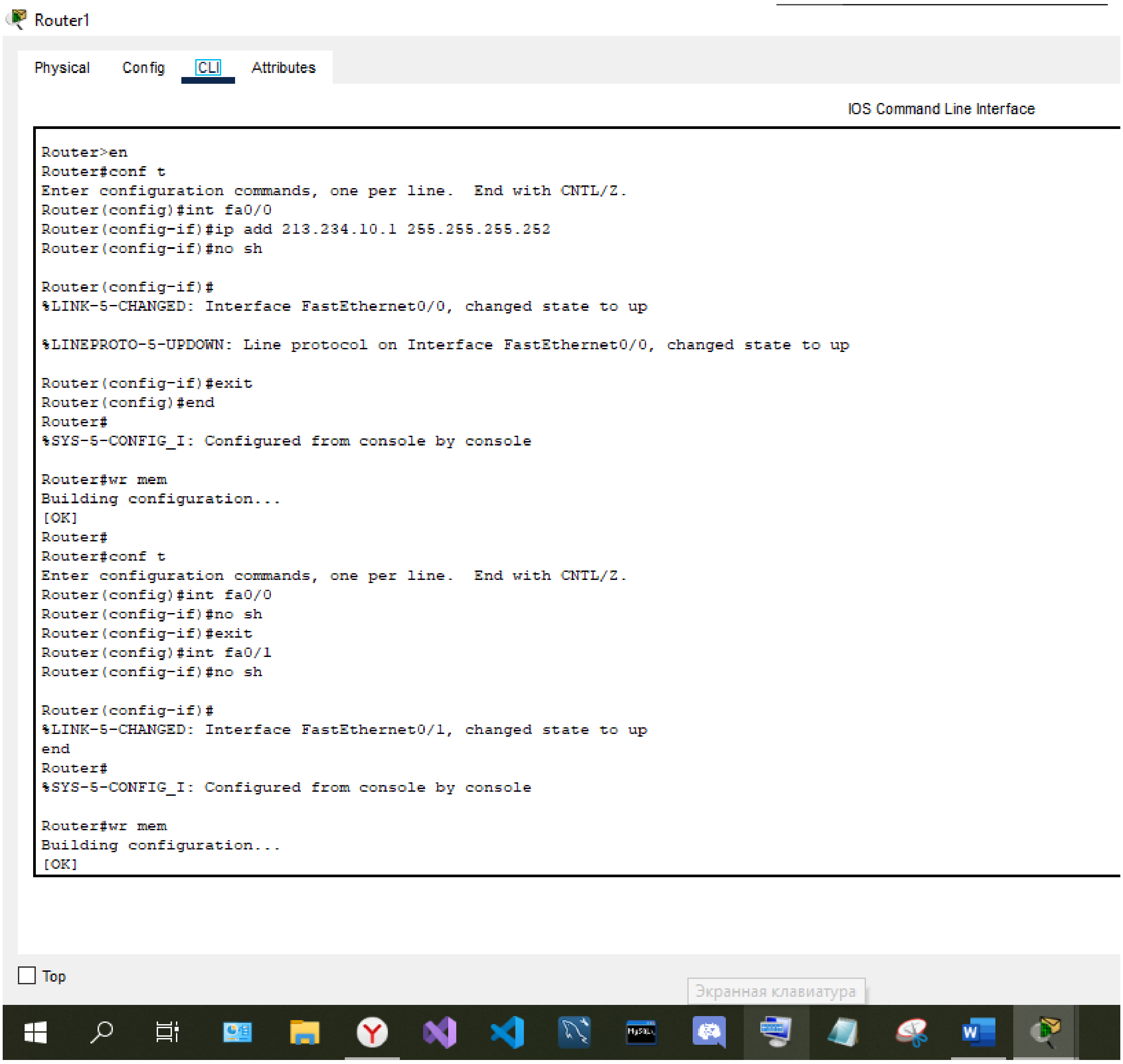
```
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up


Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name users
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name server
Switch(config-vlan)#
Switch(config-vlan)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#int range fa0/3-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

Top




```
Press RETURN to get started!



Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

Router(config-if)#int fa0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up

Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#int fa0/0.3
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up

Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
[OK]
Router#
```


Cisco Packet Tracer
File Edit Options View Tools Extensions Window
Logical
Physical
x: 310, y: 344
Fa0
Server-PT
Server0
Fa0
PC-PT
PC0
Time: 00:52:39
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.3.2
Pinging 192.168.3.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.3.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
Top
Copper Cross-Over
Toggle PDU List Window


Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 213.234.10.1 255.255.255.252
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

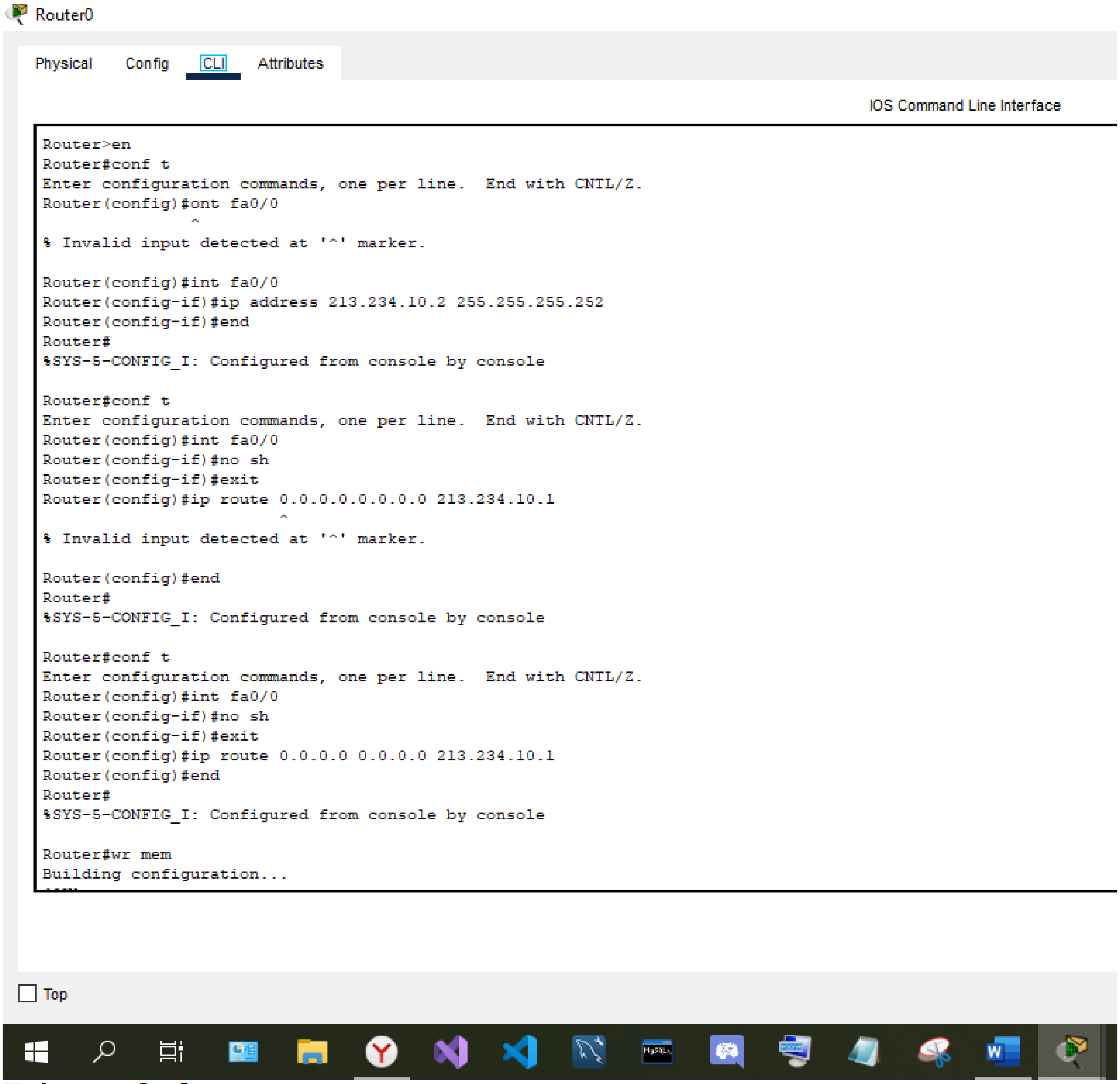
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

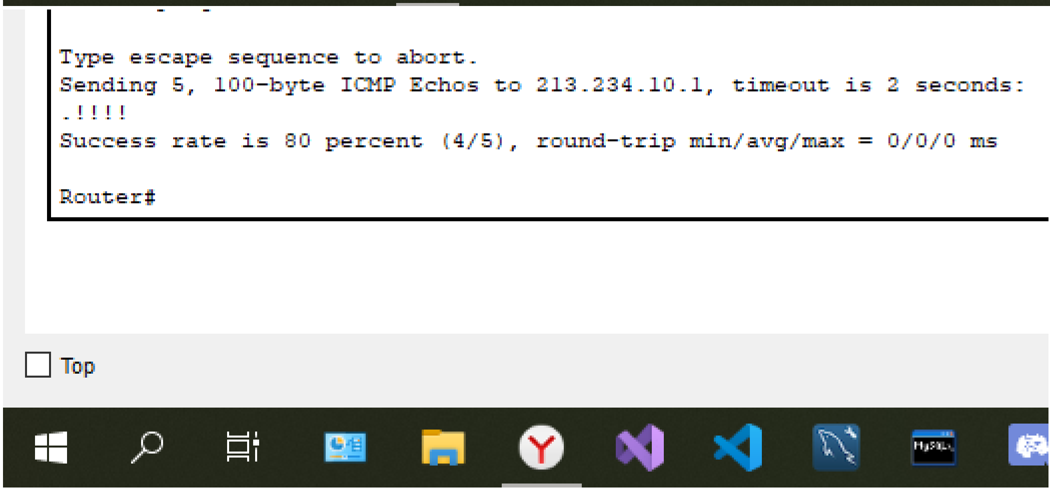
Router#wr mem
Building configuration...
[OK]
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
end
Router#
%SYS-5-CONFIG_I: Configured from console by console

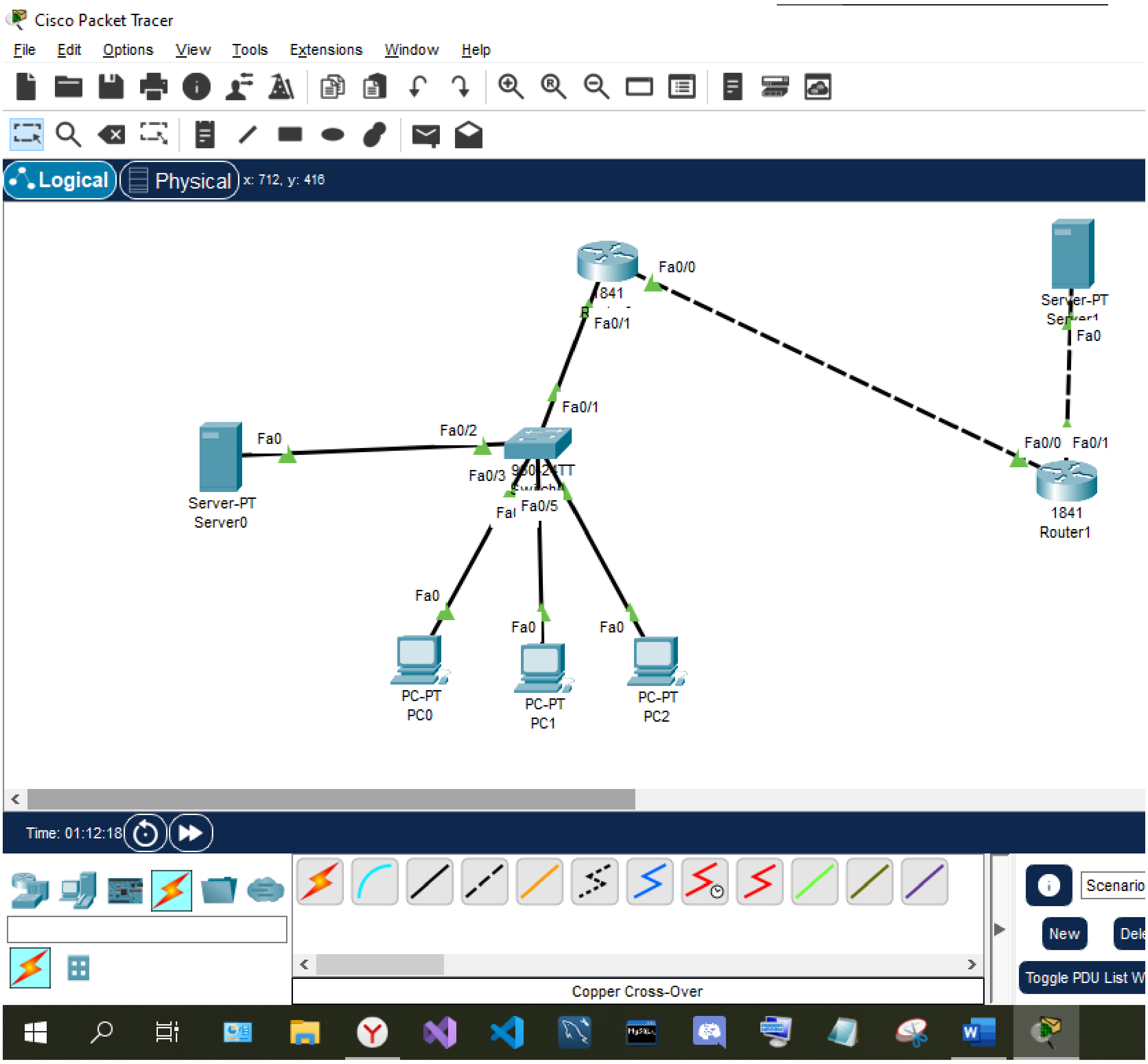
Router#wr mem
Building configuration...
[OK]
Top
Экранная клавиатура

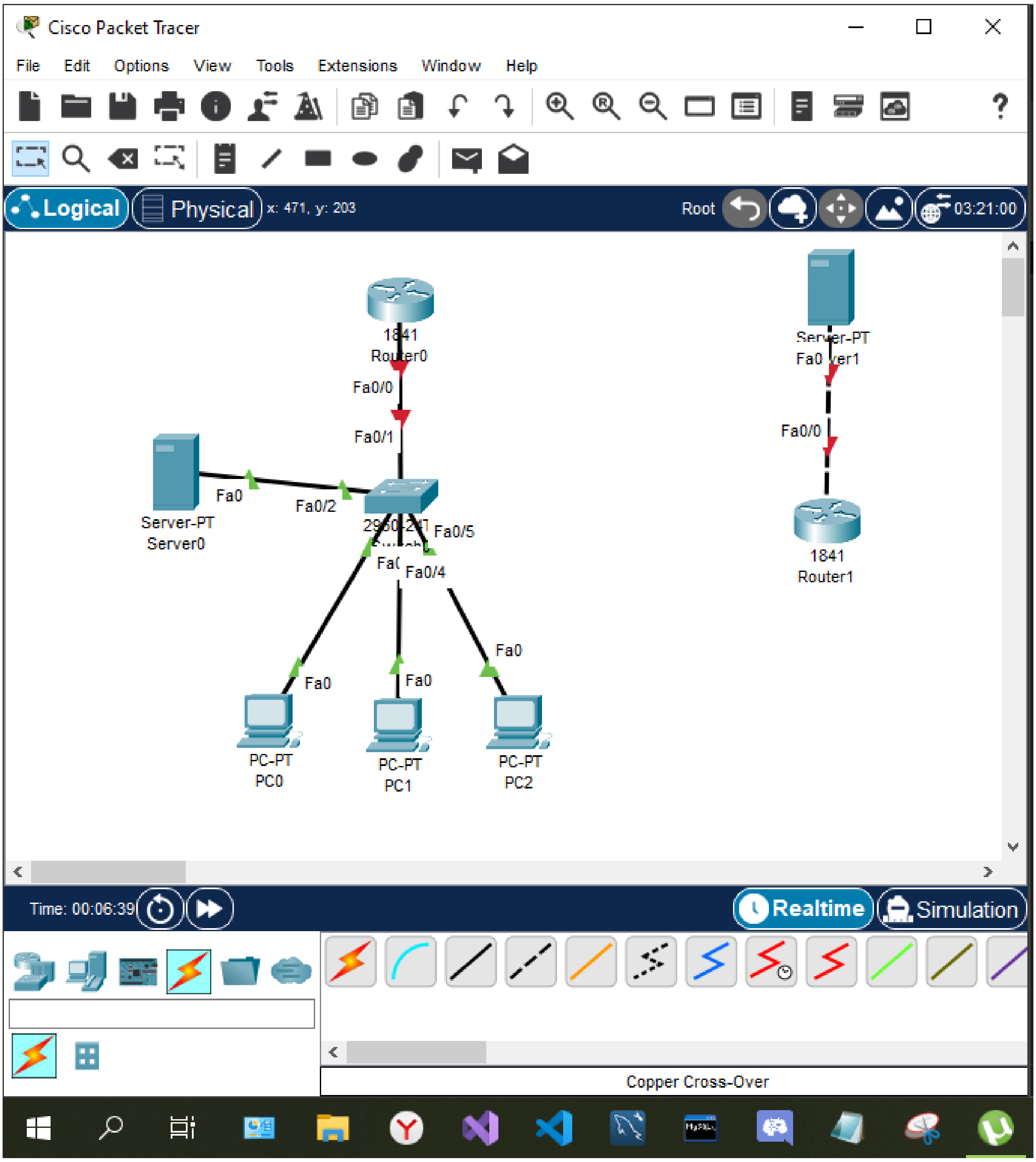



```
Router>en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ont fa0/0
                ^
% Invalid input detected at '^' marker.

Router(config)#int fa0/0
Router(config-if)#ip address 213.234.10.2 255.255.255.252
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#ip route 0.0.0.0.0.0.0.0 213.234.10.1
                        ^
% Invalid input detected at '^' marker.

Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 213.234.10.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr mem
Building configuration...
```

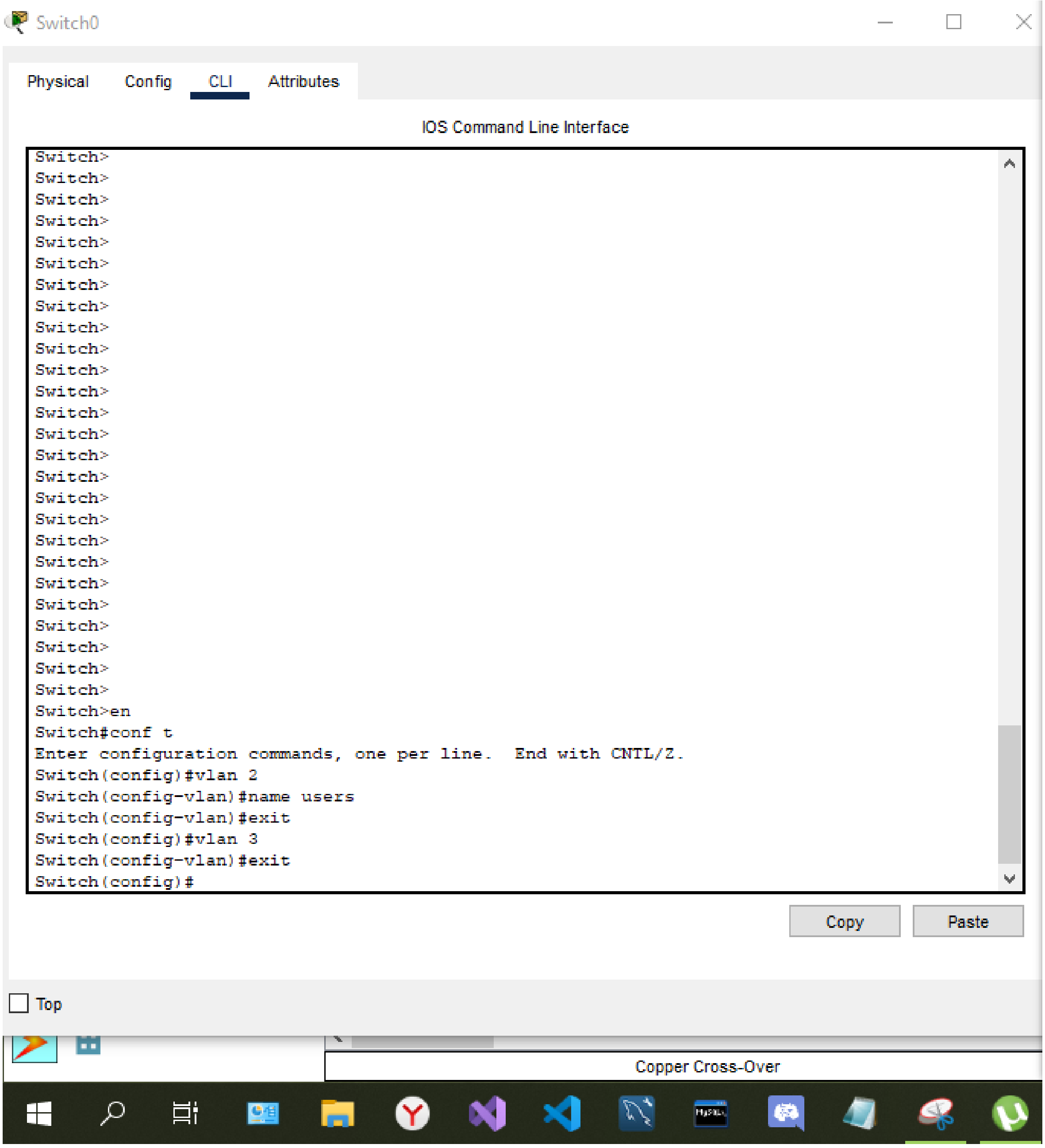


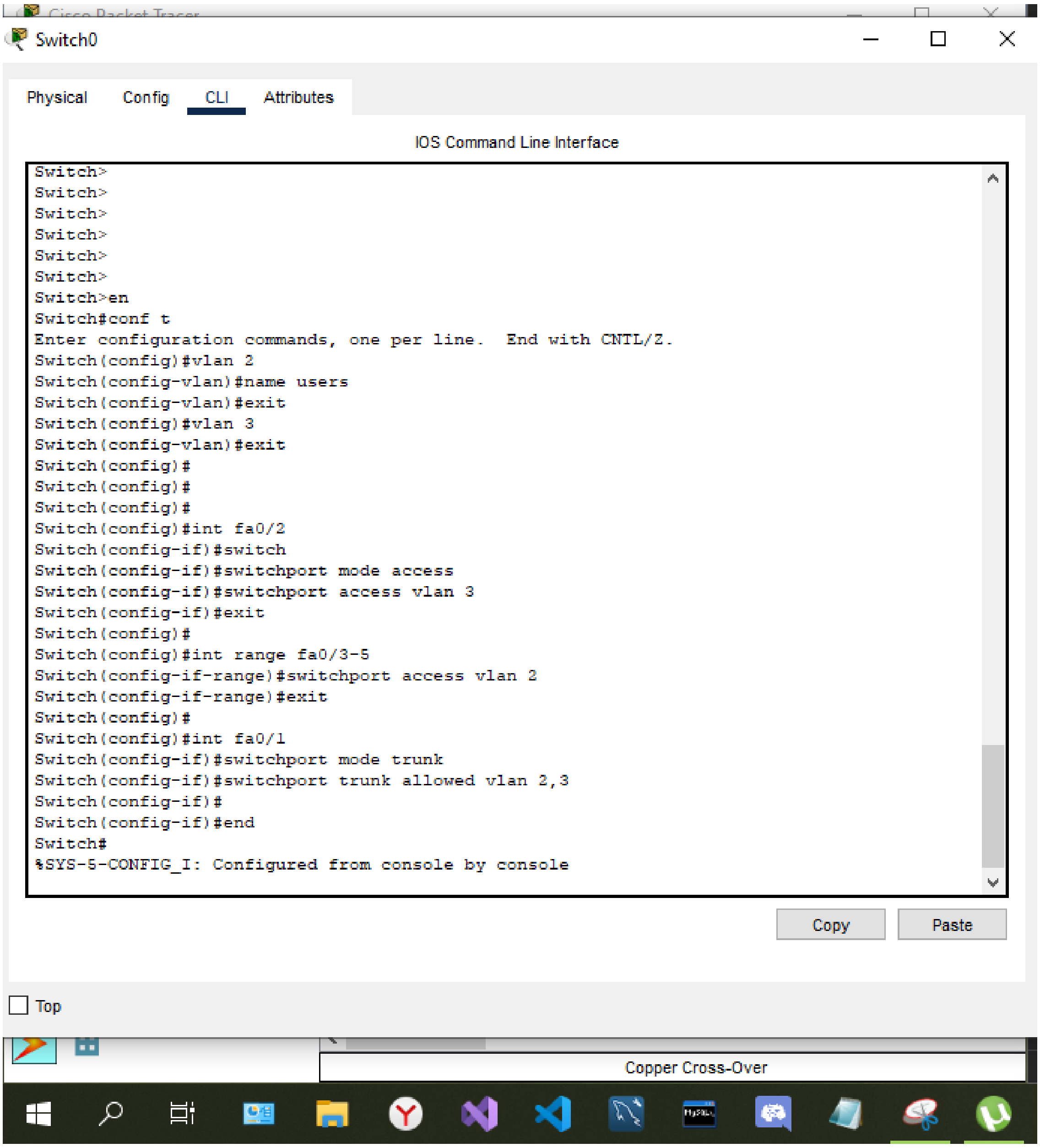

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 213.234.10.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#
```

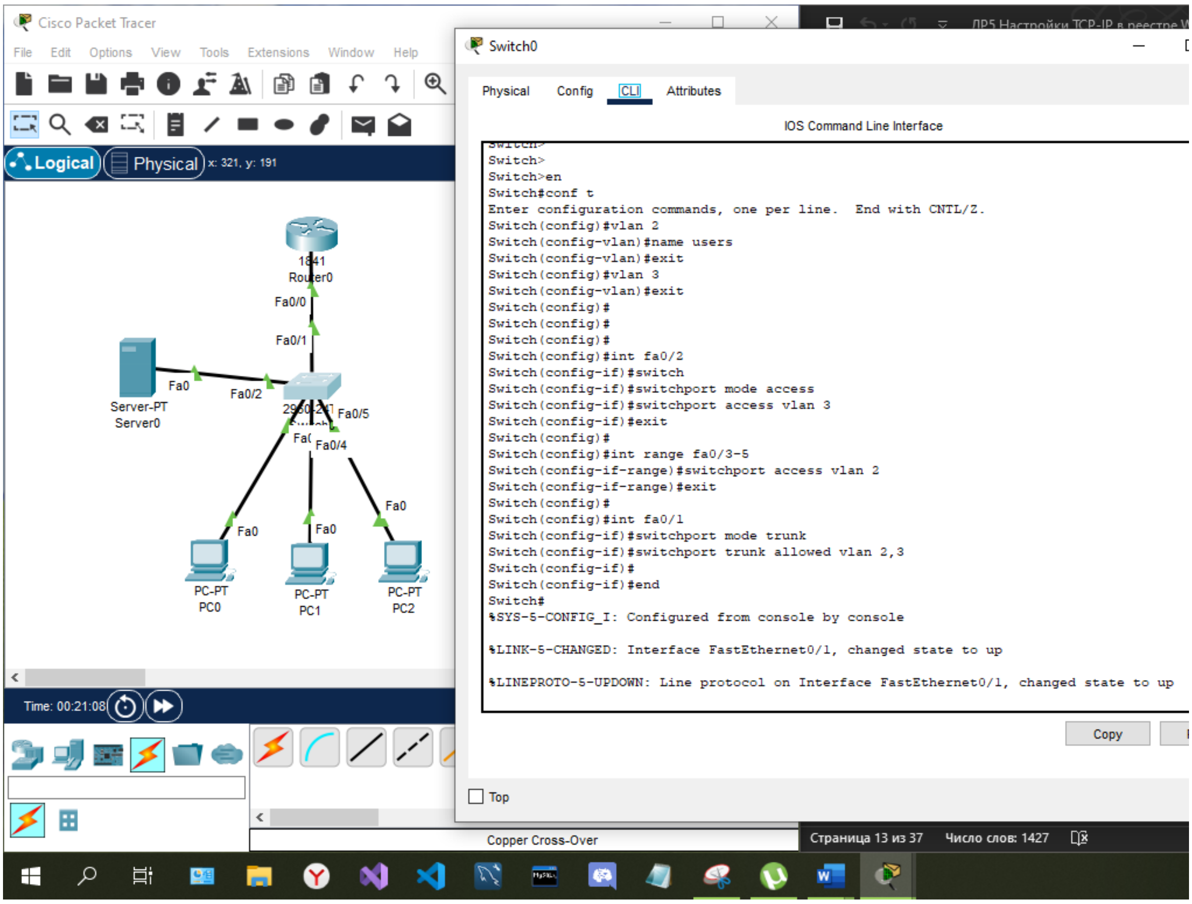

Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 712, y: 416
841
Fa0/0
Fa0/1
Server-PT
Fa0
Fa0/1
Fa0/2
Server-PT
Server0
Fa0/3
Fa0/5
Fa0/0
Fa0/1
1841
Router1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Time: 01:12:18
Scenario
New
Toggle PDU List W
Copper Cross-Over

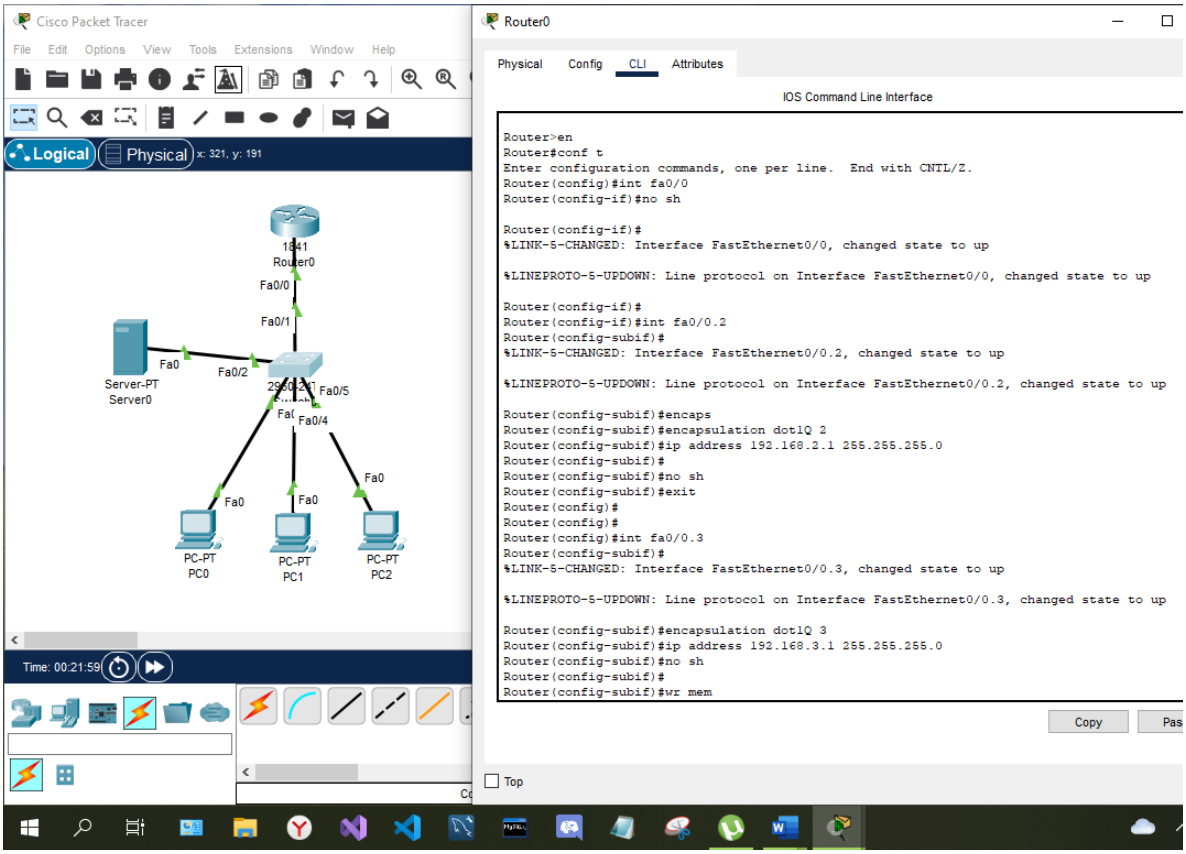


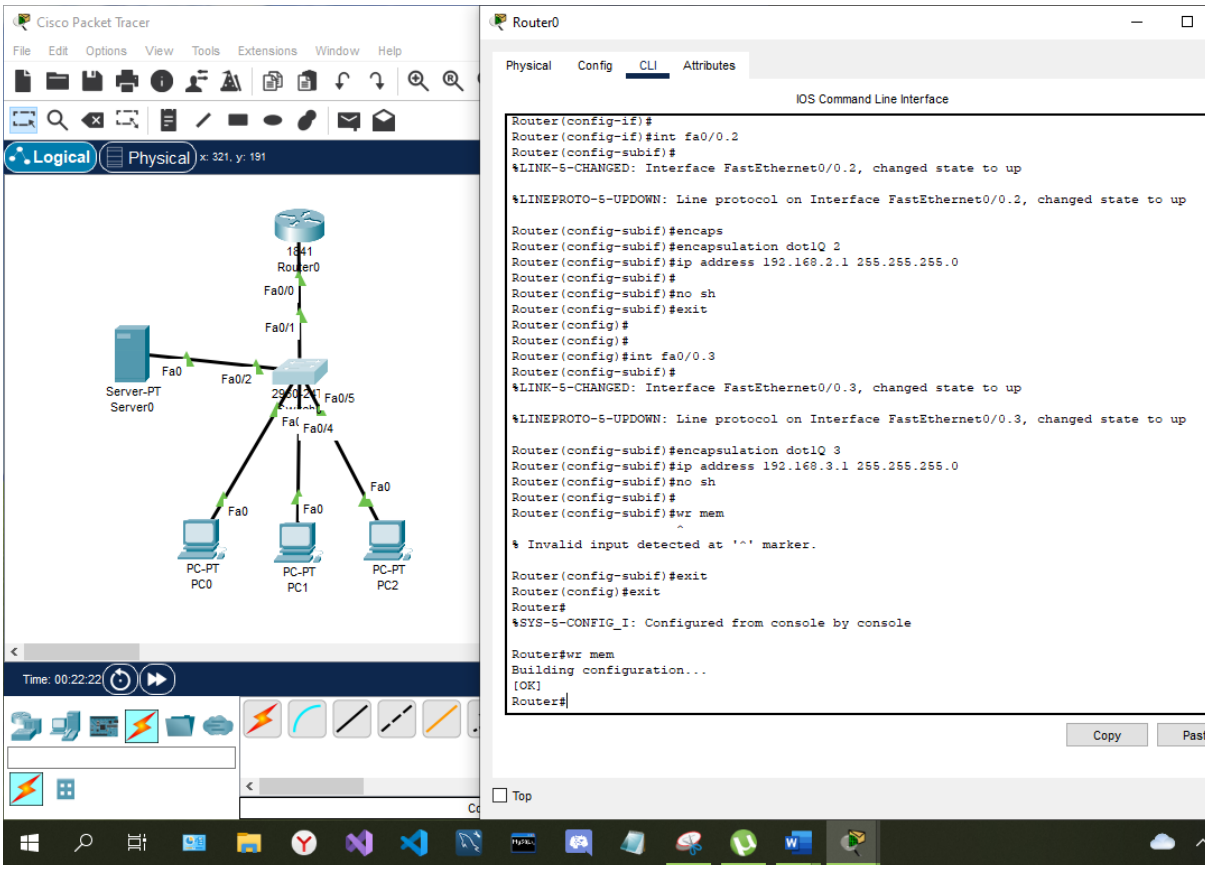
Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 471, y: 203
Root
03:21:00
1841
Router0
Fa0/0
Fa0/1
Server-PT
Server0
Fa0
Fa0/2
Fa0/5
Fa0/4
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Fa0/0
1841
Router1
Time: 00:06:39
Realtime
Simulation
Copper Cross-Over

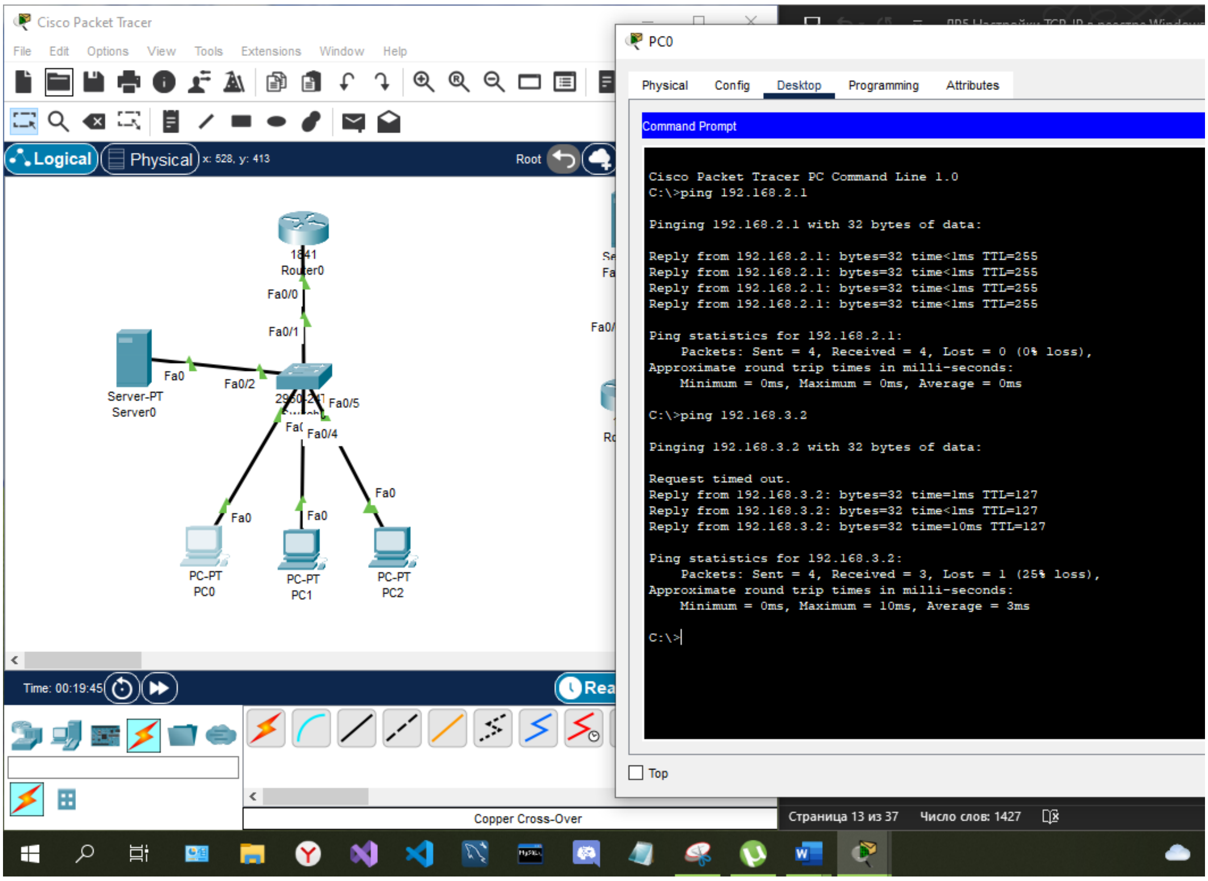

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name users
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#exit
Switch(config)#
Copy
Paste
Top
Copper Cross-Over

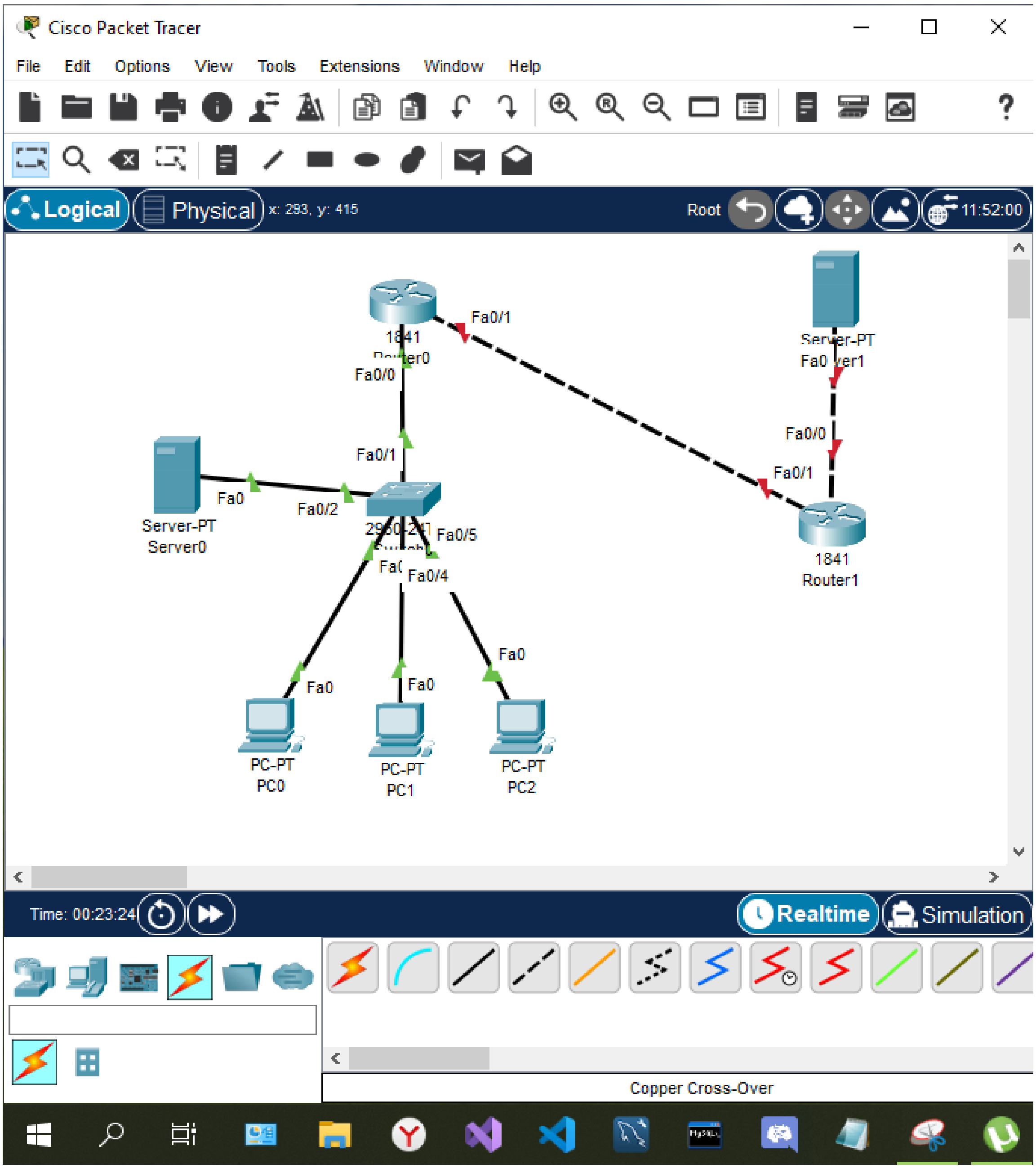

Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name users
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#int fa0/2
Switch(config-if)#switch
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#
Switch(config)#int range fa0/3-5
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top
Copper Cross-Over


Cisco Packet Tracer
Switch0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Switch>
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name users
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#exit
Switch(config)#
Switch(config)#
Switch(config)#
Switch(config)#int fa0/2
Switch(config-if)#switch
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#
Switch(config)#int range fa0/3-5
Switch(config-if-range)#switchport access vlan 2
Switch(config-if-range)#exit
Switch(config)#
Switch(config)#int fa0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Copy
Top
1841
Router0
Fa0/0
Fa0/1
Server-PT
Server0
Fa0
Fa0/2
Fa0/5
Fa0/4
PC-PT
PC0
PC1
PC2
Time: 00:21:08
Copper Cross-Over
Страница 13 из 37
Число слов: 1427

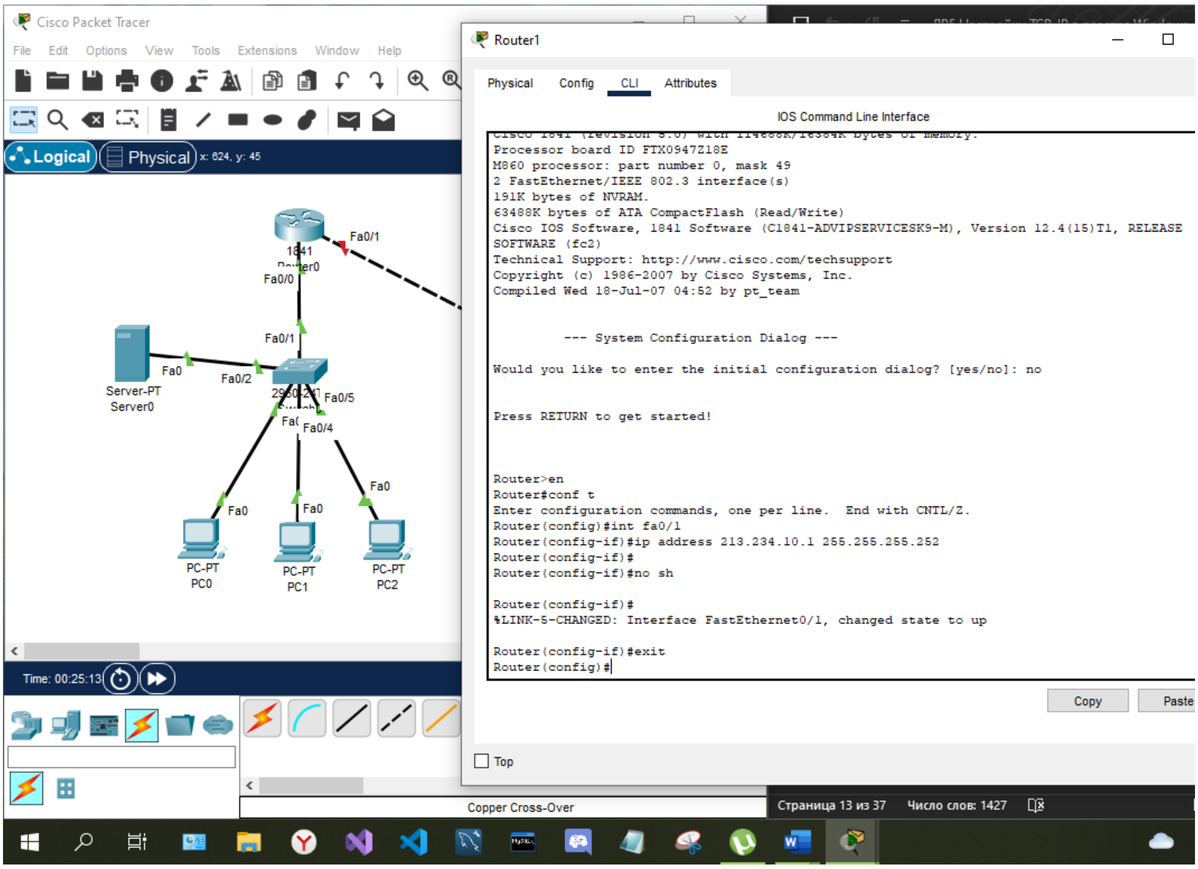

Cisco Packet Tracer
Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#
Router(config-if)#int fa0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
Router(config-subif)#encaps
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#
Router(config)#
Router(config)#int fa0/0.3
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#
Router(config-subif)#wr mem
Copy
Top
1841
Router0
Fa0/0
Fa0/1
Server-PT
Server0
Fa0
Fa0/2
Fa0/5
Fa0/4
PC-PT
PC0
PC1
PC2
Time: 00:21:59

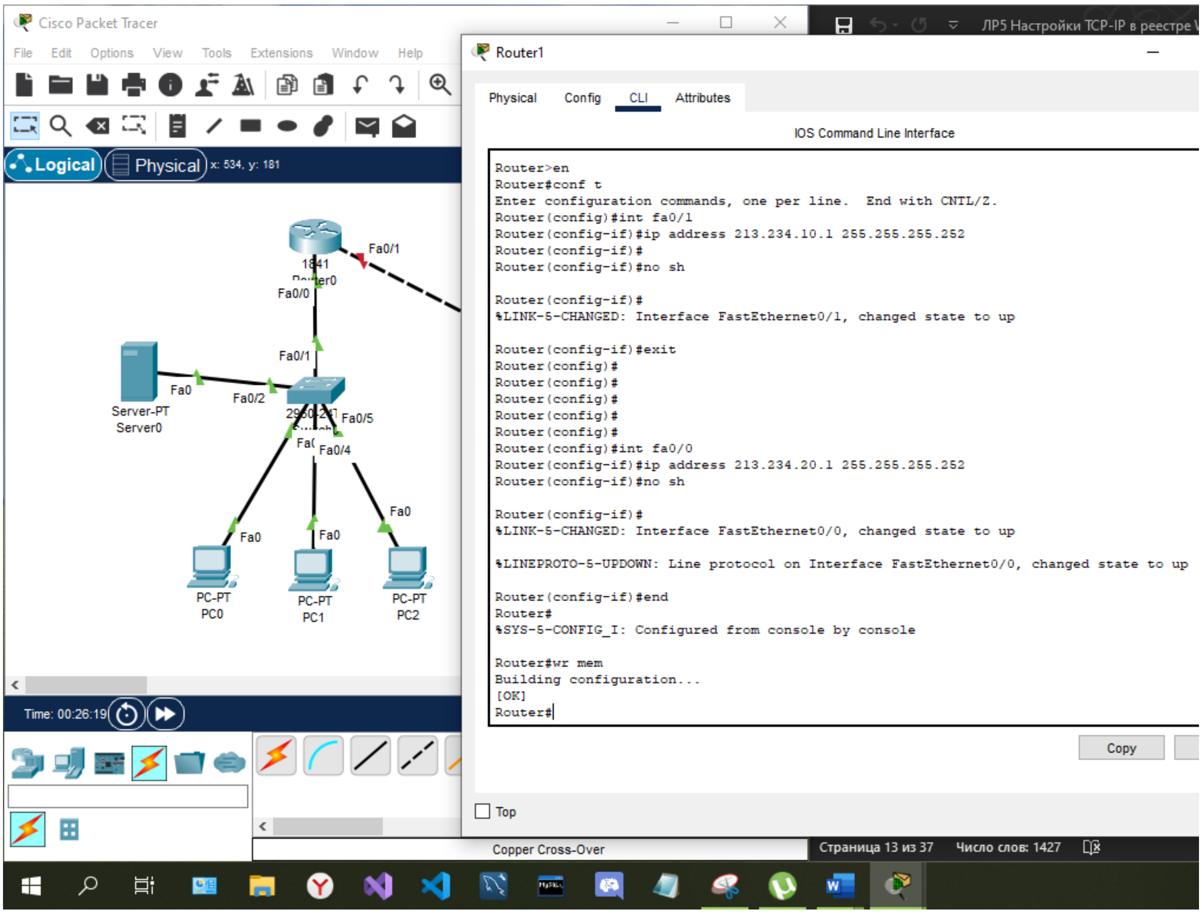

Cisco Packet Tracer
Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config-if)#
Router(config-if)#int fa0/0.2
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.2, changed state to up
Router(config-subif)#encaps
Router(config-subif)#encapsulation dot1Q 2
Router(config-subif)#ip address 192.168.2.1 255.255.255.0
Router(config-subif)#
Router(config-subif)#no sh
Router(config-subif)#exit
Router(config)#
Router(config)#
Router(config)#int fa0/0.3
Router(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.3, changed state to up
Router(config-subif)#encapsulation dot1Q 3
Router(config-subif)#ip address 192.168.3.1 255.255.255.0
Router(config-subif)#no sh
Router(config-subif)#
Router(config-subif)#wr mem
% Invalid input detected at '^' marker.
Router(config-subif)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Copy
Top
Time: 00:22:22

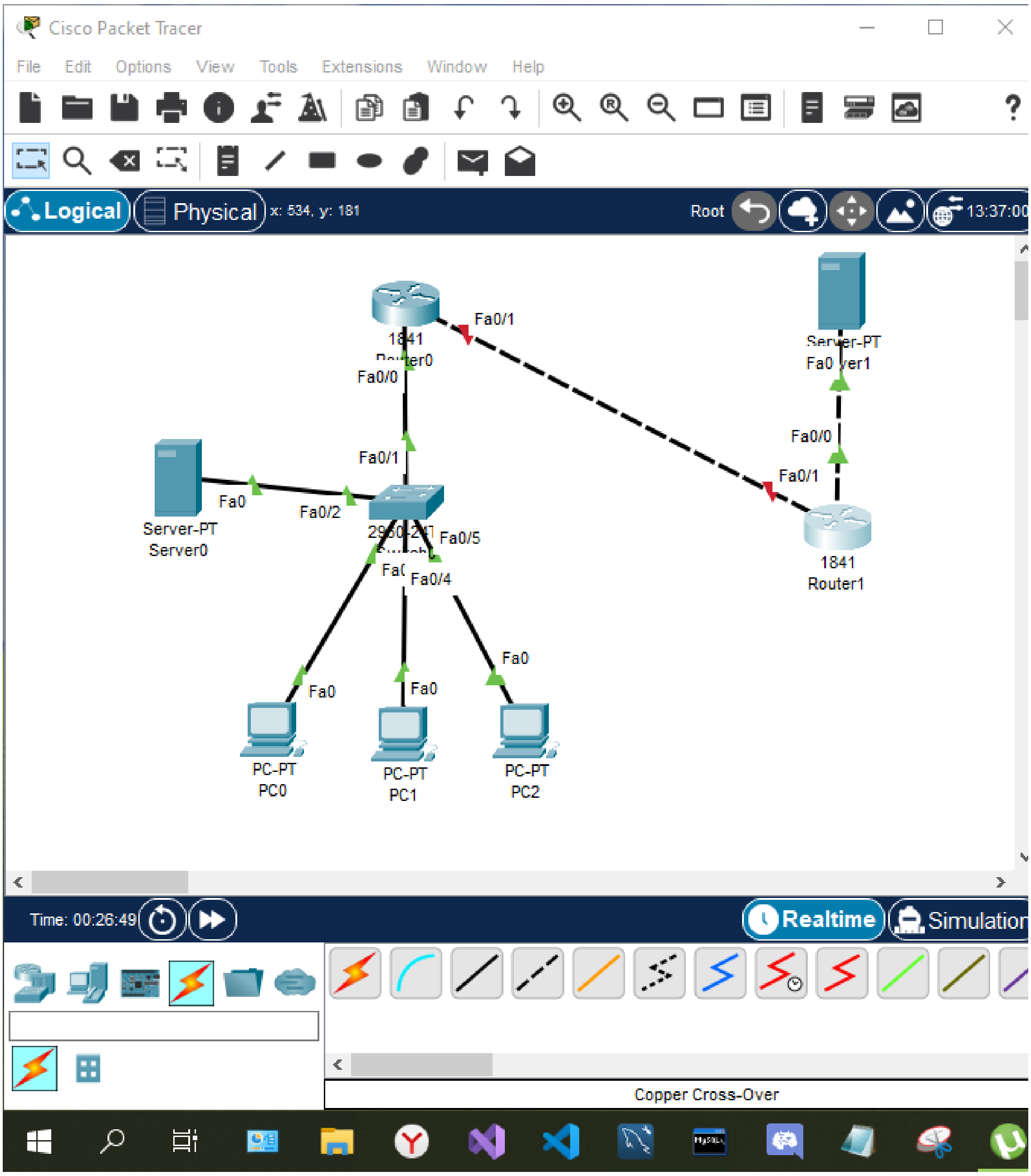

Cisco Packet Tracer
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.1
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.3.2
Pinging 192.168.3.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=10ms TTL=127
Ping statistics for 192.168.3.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms
C:\>
Top
Time: 00:19:45
Copper Cross-Over
Страница 13 из 37
Число слов: 1427

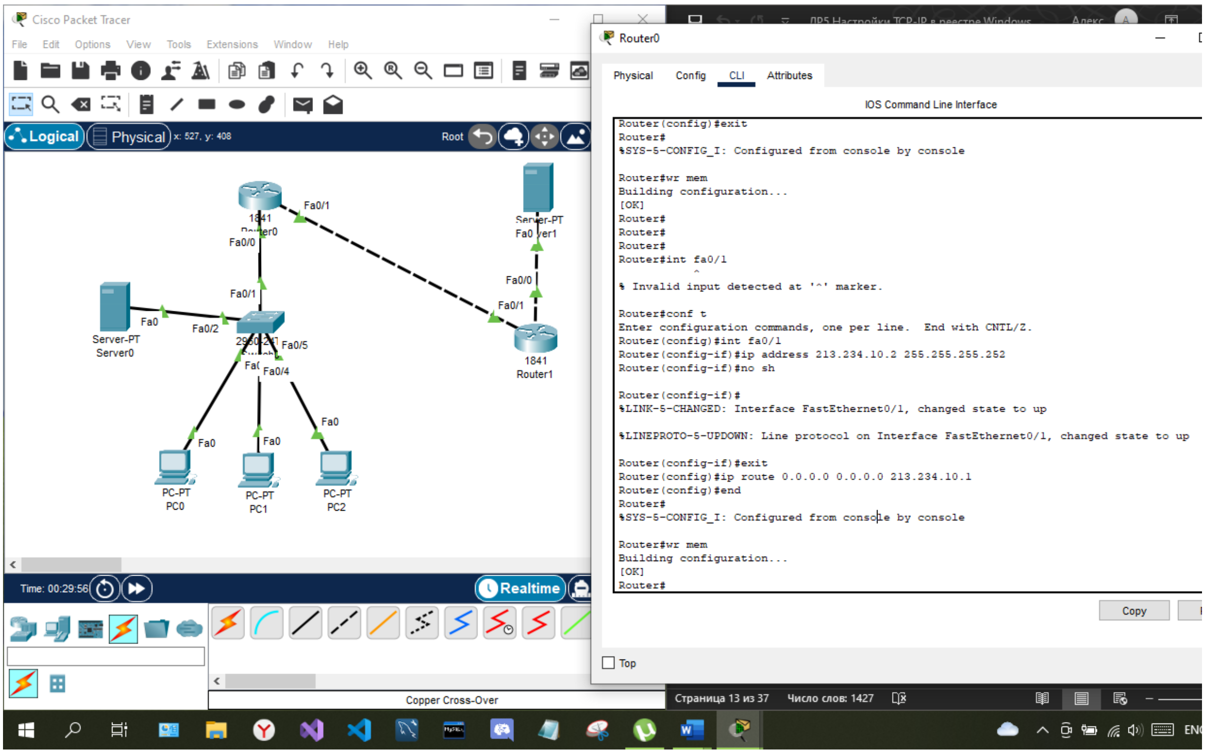


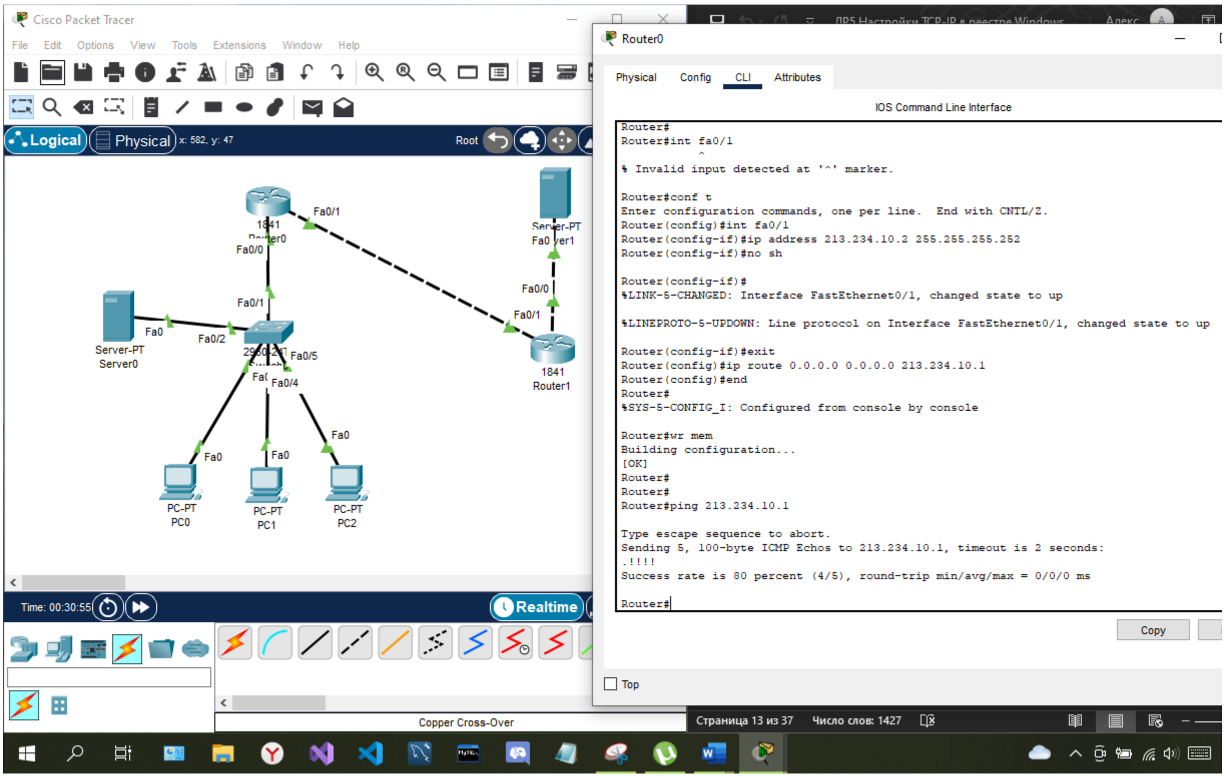
Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 293, y: 415
Root
11:52:00
Fa0/1
1841
Router0
Fa0/0
Fa0/1
Server-PT
Server0
Fa0
Fa0/2
Fa0/5
Fa0/4
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Server-PT
Server1
Fa0
Fa0/0
Fa0/1
1841
Router1
Time: 00:23:24
Realtime
Simulation
Copper Cross-Over

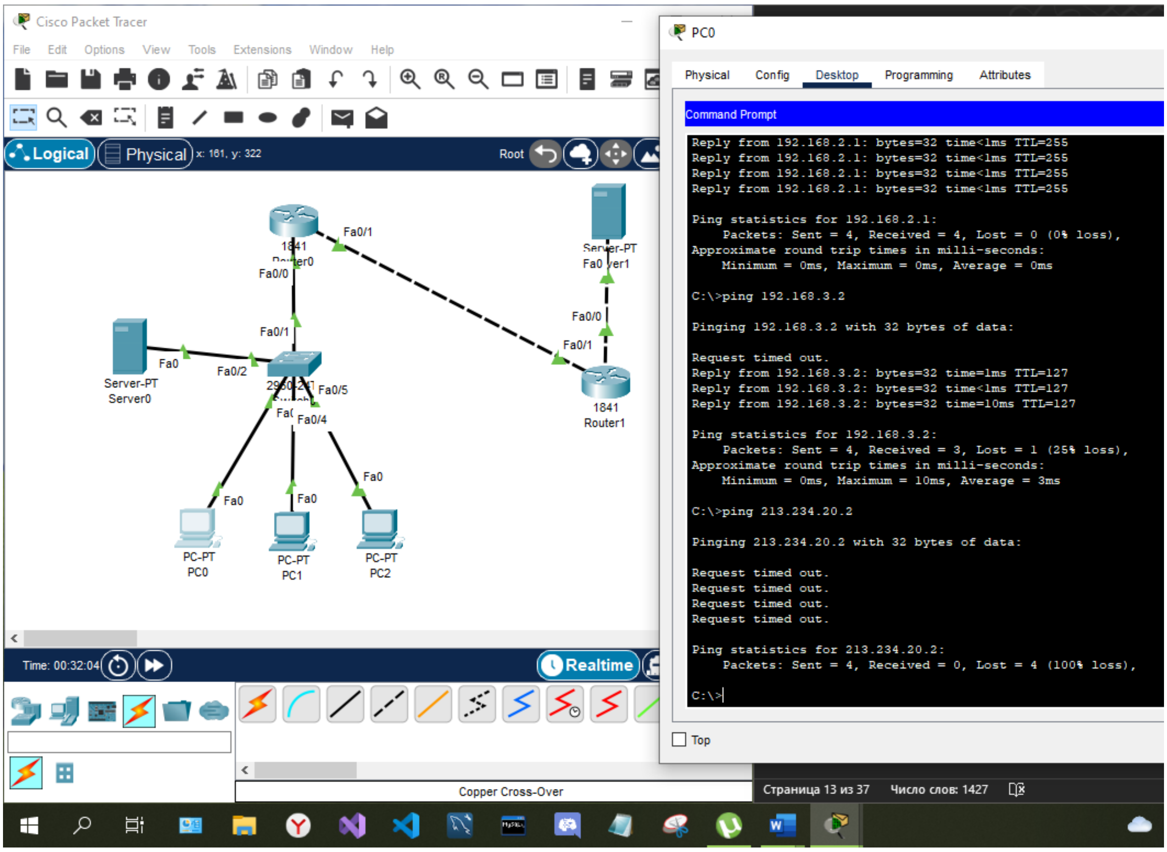

Cisco Packet Tracer
Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Processor board ID FTX0947Z18E
M860 processor: part number 0, mask 49
2 FastEthernet/IEEE 802.3 interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team
--- System Configuration Dialog ---
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip address 213.234.10.1 255.255.255.252
Router(config-if)#
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
Copy
Paste
Top
Server-PT
Server0
PC-PT
PC0
PC1
PC2
Time: 00:25:13
Copper Cross-Over
Страница 13 из 37
Число слов: 1427

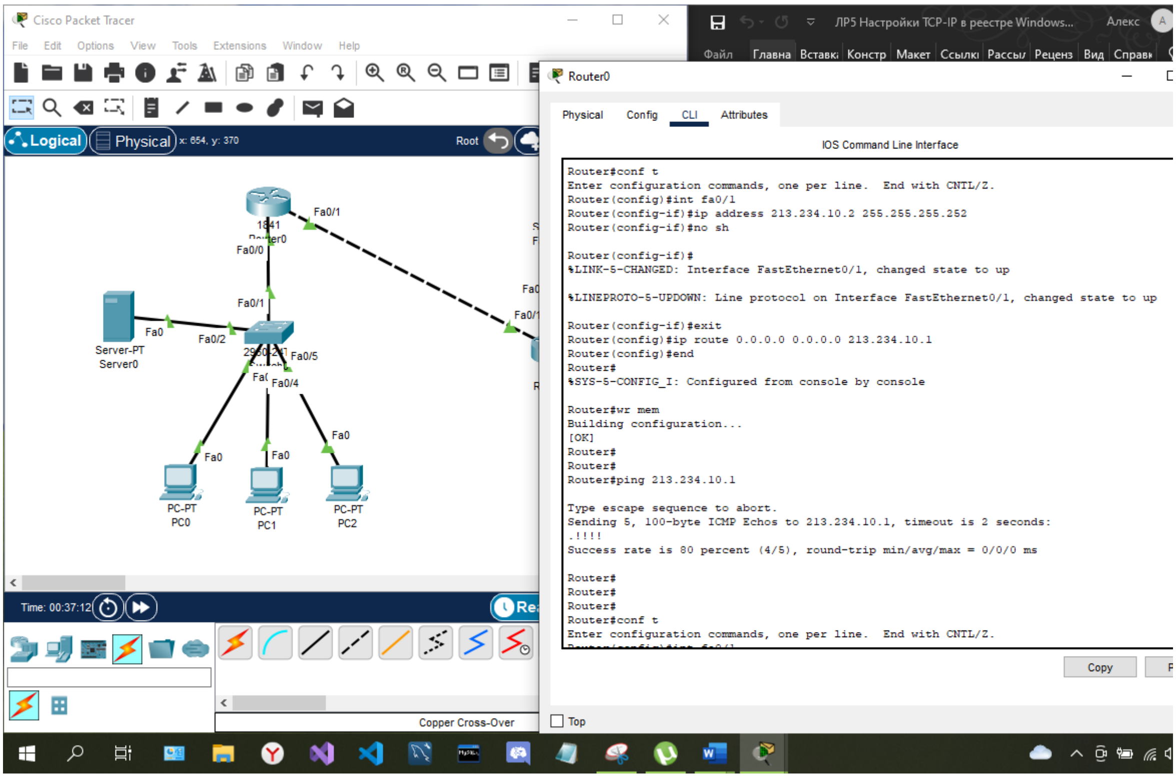

Cisco Packet Tracer
Router1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip address 213.234.10.1 255.255.255.252
Router(config-if)#
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#
Router(config)#int fa0/0
Router(config-if)#ip address 213.234.20.1 255.255.255.252
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Copy
Top
Server-PT
Server0
PC-PT
PC0
PC1
PC2
Time: 00:26:19
Copper Cross-Over
ЛР5 Настройки TCP-IP в реестре
Страница 13 из 37
Число слов: 1427

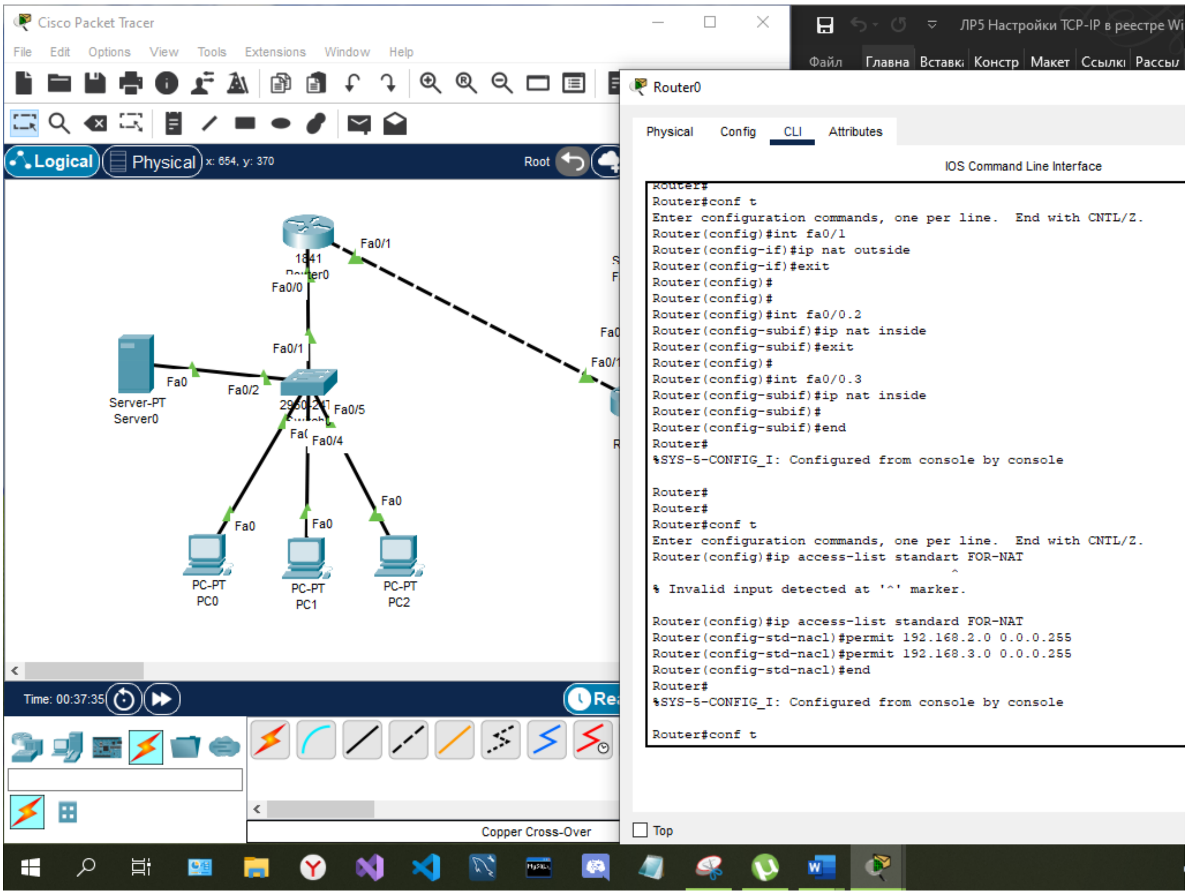

Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 534, y: 181
Root
13:37:00
Fa0/1
1841
Fa0/0
Server-PT
Fa0/0
Fa0/1
Fa0
Fa0/2
Server-PT
Server0
Fa0/5
Fa0/4
1841
Router1
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Time: 00:26:49
Realtime
Copper Cross-Over

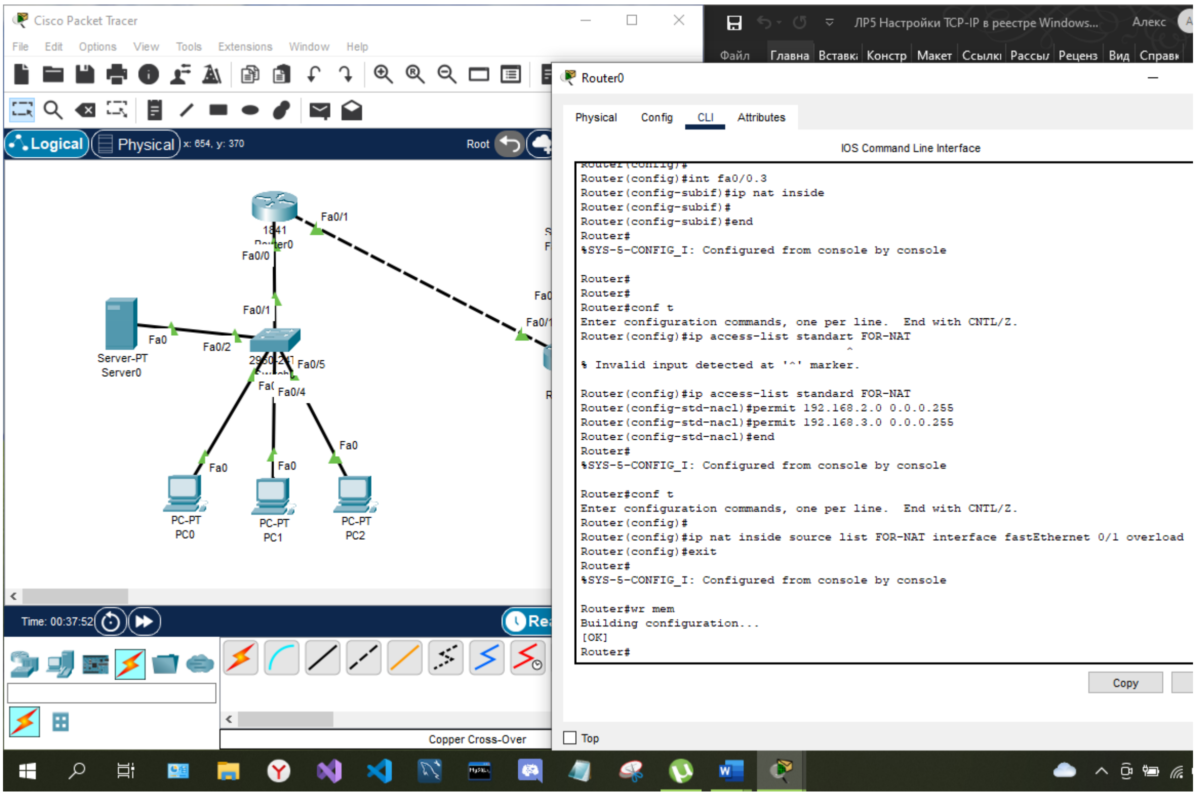

Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Router#
Router#
Router#int fa0/1
% Invalid input detected at '^' marker.
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip address 213.234.10.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 213.234.10.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Copy
Top

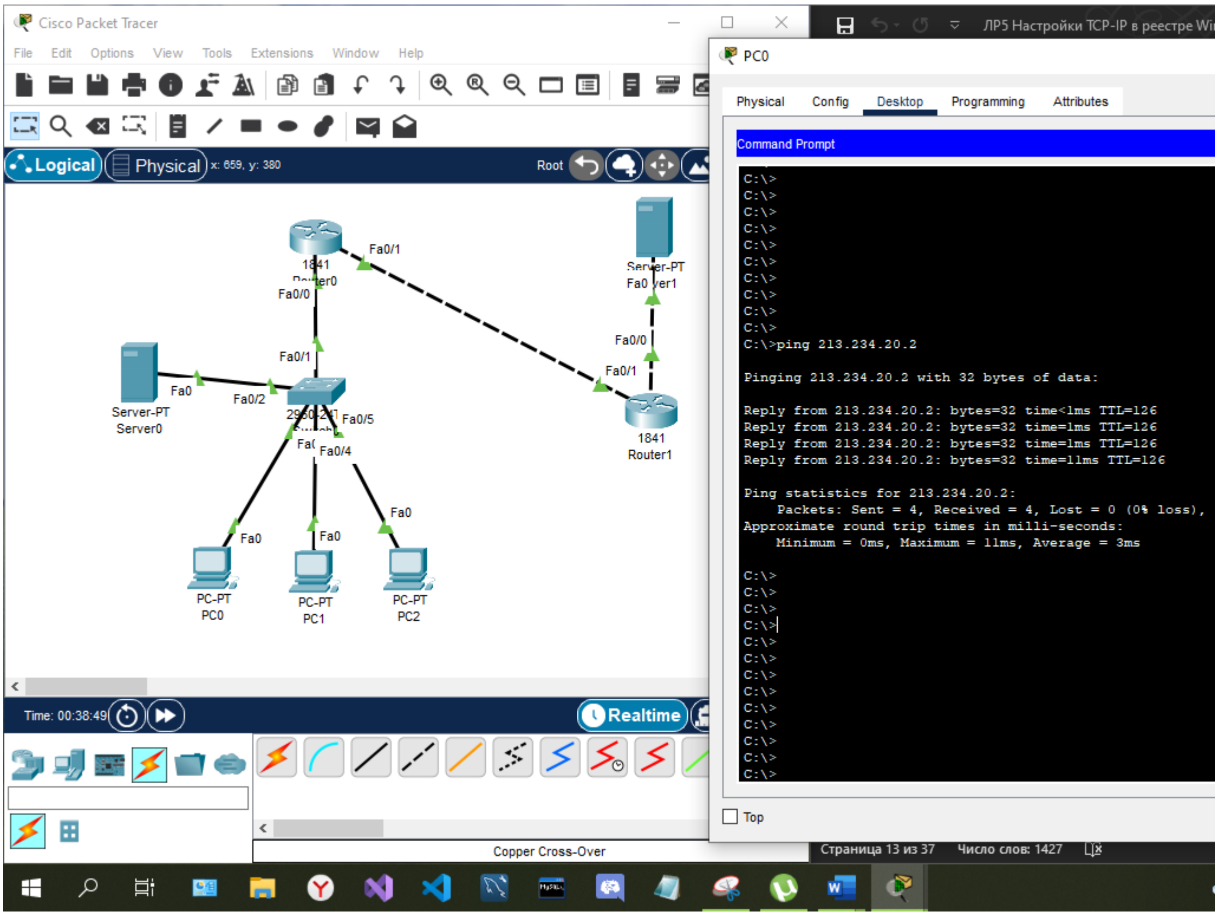

Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#
Router#int fa0/1
% Invalid input detected at '^' marker.
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip address 213.234.10.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 213.234.10.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Router#
Router#ping 213.234.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 213.234.10.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
Router#
Copy
Top


PC0
Command Prompt
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.3.2
Pinging 192.168.3.2 with 32 bytes of data:
Request timed out.
Reply from 192.168.3.2: bytes=32 time=1ms TTL=127
Reply from 192.168.3.2: bytes=32 time<1ms TTL=127
Reply from 192.168.3.2: bytes=32 time=10ms TTL=127
Ping statistics for 192.168.3.2:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 10ms, Average = 3ms
C:\>ping 213.234.20.2
Pinging 213.234.20.2 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.
Ping statistics for 213.234.20.2:
Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>


Router0
IOS Command Line Interface
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip address 213.234.10.2 255.255.255.252
Router(config-if)#no sh
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 213.234.10.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Router#
Router#ping 213.234.10.1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 213.234.10.1, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms
Router#
Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.


Router0
IOS Command Line Interface
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#int fa0/0.2
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#
Router(config)#int fa0/0.3
Router(config-subif)#ip nat inside
Router(config-subif)#
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list standart FOR-NAT
% Invalid input detected at '^' marker.
Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.2.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.3.0 0.0.0.255
Router(config-std-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#conf t

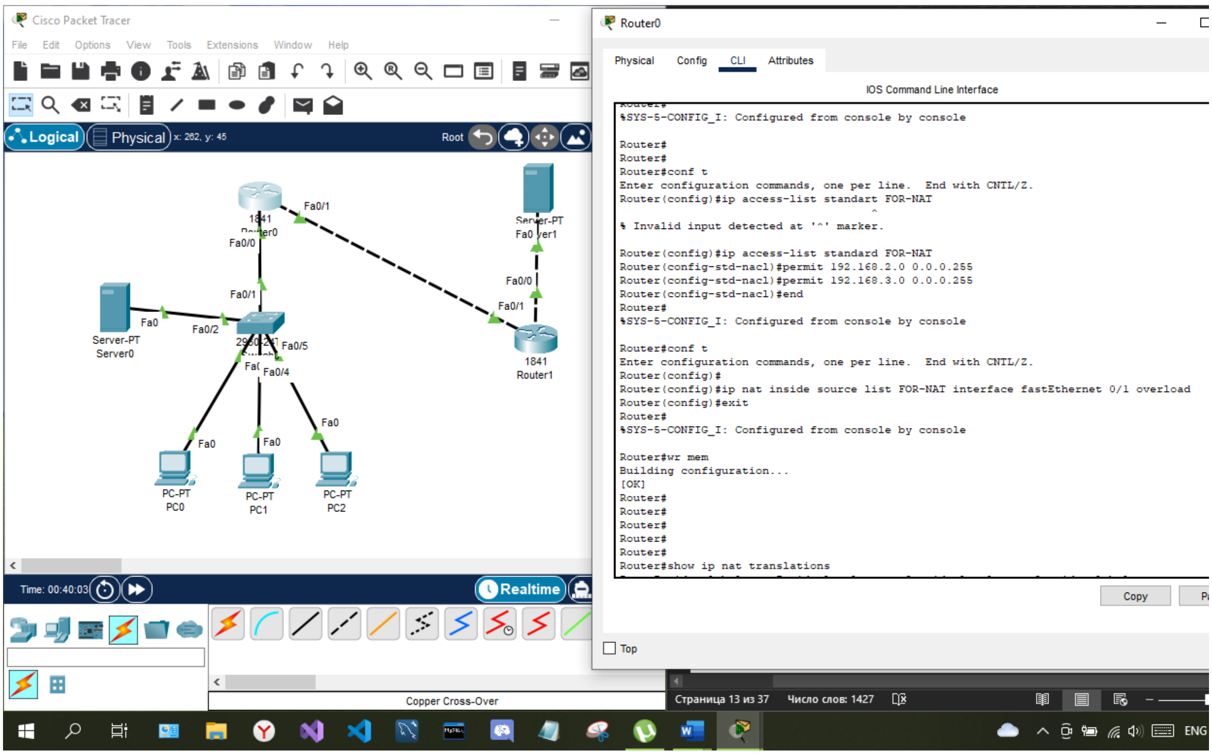

Router0
IOS Command Line Interface
Router(config)#int fa0/0.3
Router(config-subif)#ip nat inside
Router(config-subif)#
Router(config-subif)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list standart FOR-NAT
% Invalid input detected at '^' marker.
Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.2.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.3.0 0.0.0.255
Router(config-std-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip nat inside source list FOR-NAT interface fastEthernet 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#


Cisco Packet Tracer
ЛР5 Настройки TCP-IP в реестре Wi
PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
C:\>ping 213.234.20.2

Pinging 213.234.20.2 with 32 bytes of data:

Reply from 213.234.20.2: bytes=32 time<1ms TTL=126
Reply from 213.234.20.2: bytes=32 time=1ms TTL=126
Reply from 213.234.20.2: bytes=32 time=1ms TTL=126
Reply from 213.234.20.2: bytes=32 time=11ms TTL=126

Ping statistics for 213.234.20.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 11ms, Average = 3ms
Logical
Physical
Root
1841
Fa0/1
Fa0/0
Server-PT
Server0
Fa0/2
Fa0/5
Fa0/4
Fa0
PC-PT
PC0
PC1
PC2
1841
Router1
Time: 00:38:49
Realtime
Copper Cross-Over
Top
Страница 13 из 37
Число слов: 1427

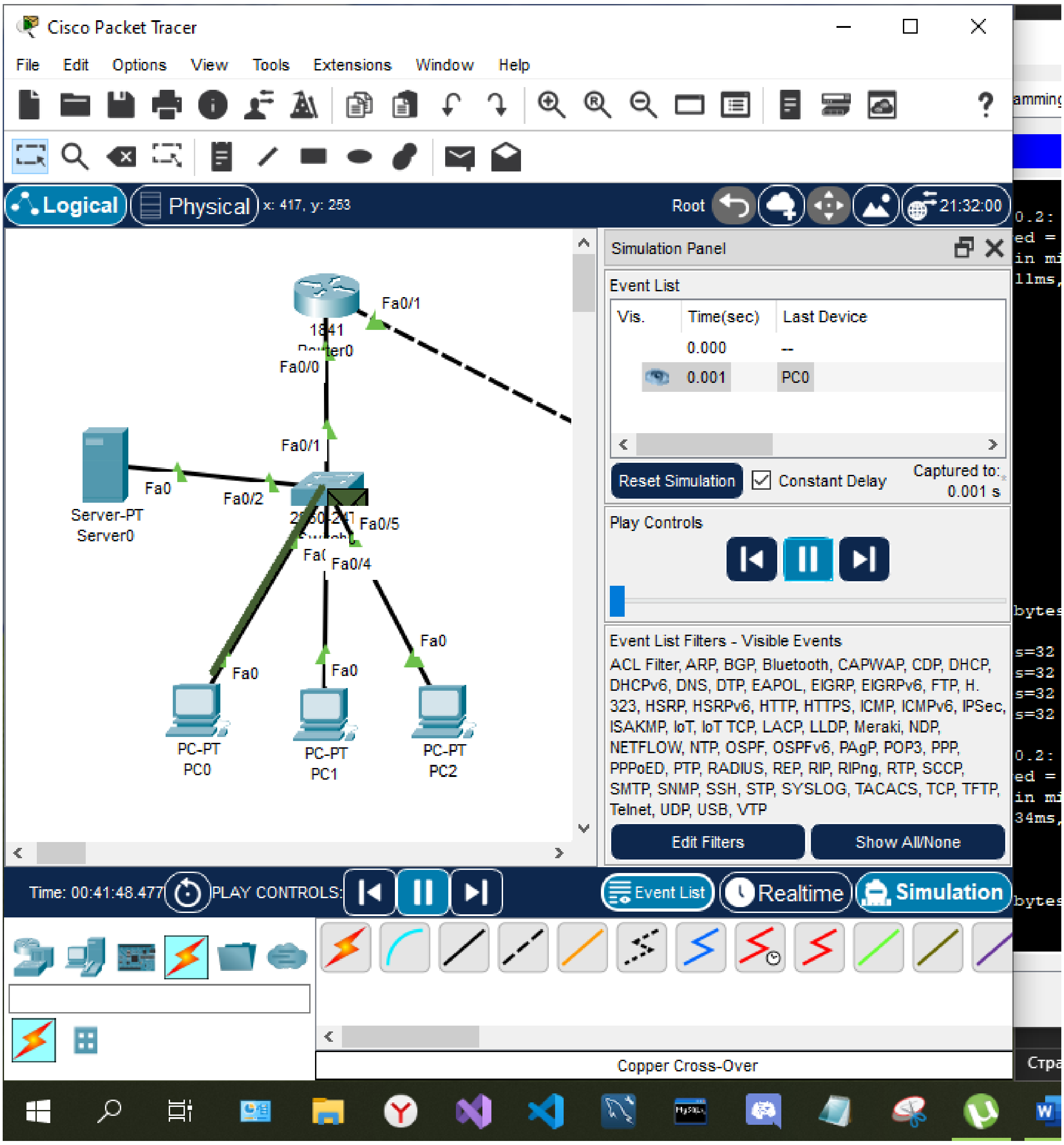

Cisco Packet Tracer
Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip access-list standart FOR-NAT
^
% Invalid input detected at '^' marker.

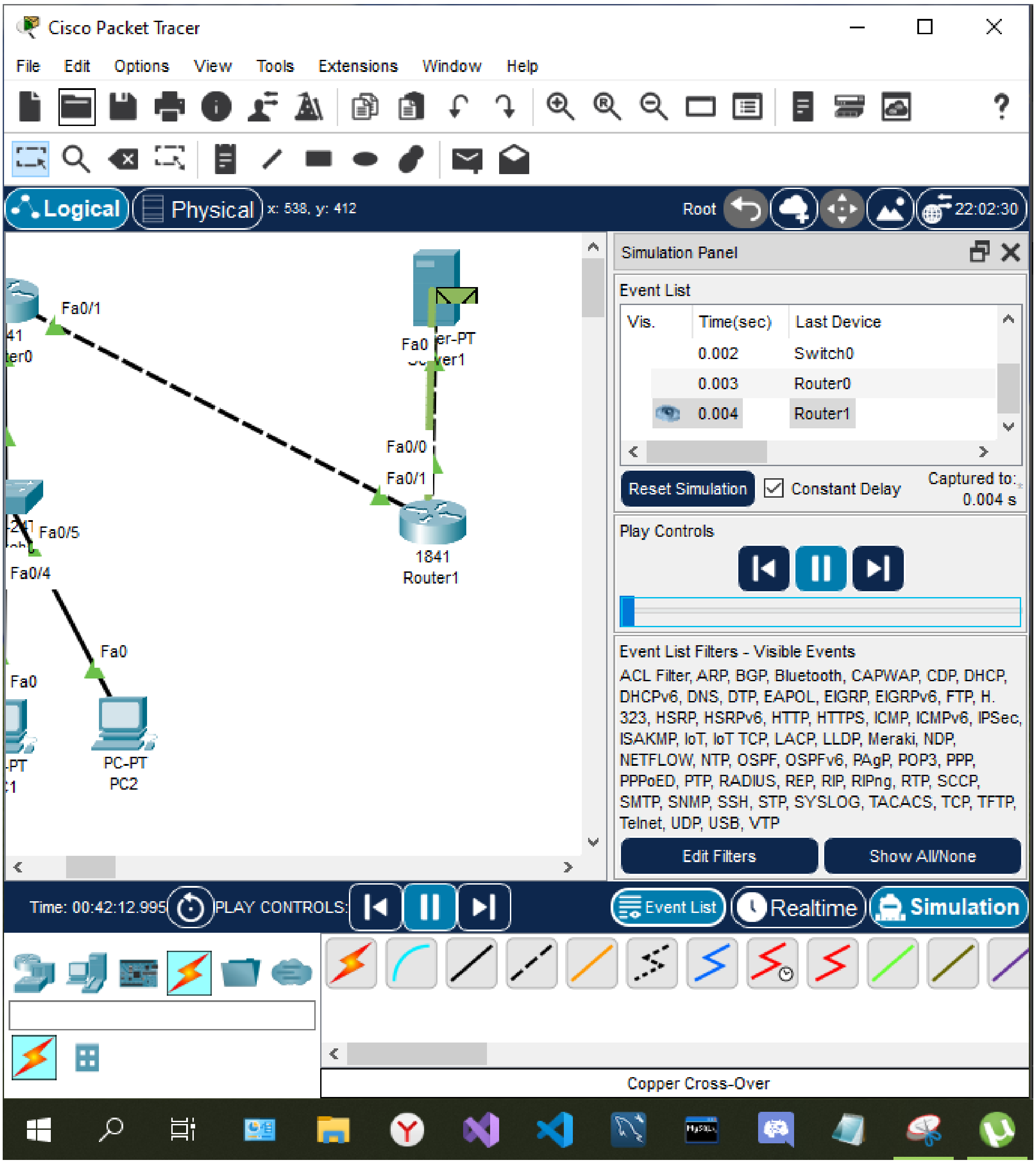
Router(config)#ip access-list standard FOR-NAT
Router(config-std-nacl)#permit 192.168.2.0 0.0.0.255
Router(config-std-nacl)#permit 192.168.3.0 0.0.0.255
Router(config-std-nacl)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip nat inside source list FOR-NAT interface fastEthernet 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

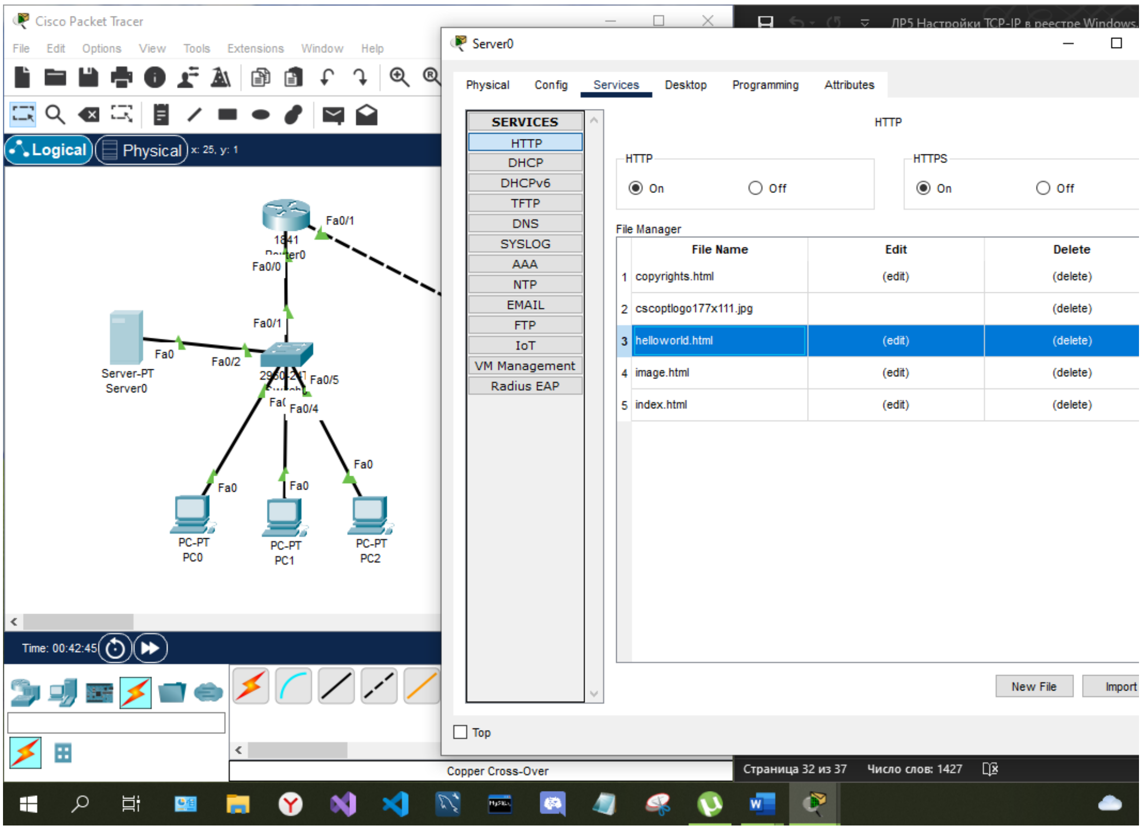
Router#wr mem
Building configuration...
[OK]
Router#
Router#
Router#
Router#
Router#
Router#show ip nat translations
Copy
Top
Time: 00:40:03
Realtime
Copper Cross-Over
Страница 13 из 37
Число слов: 1427
ENG

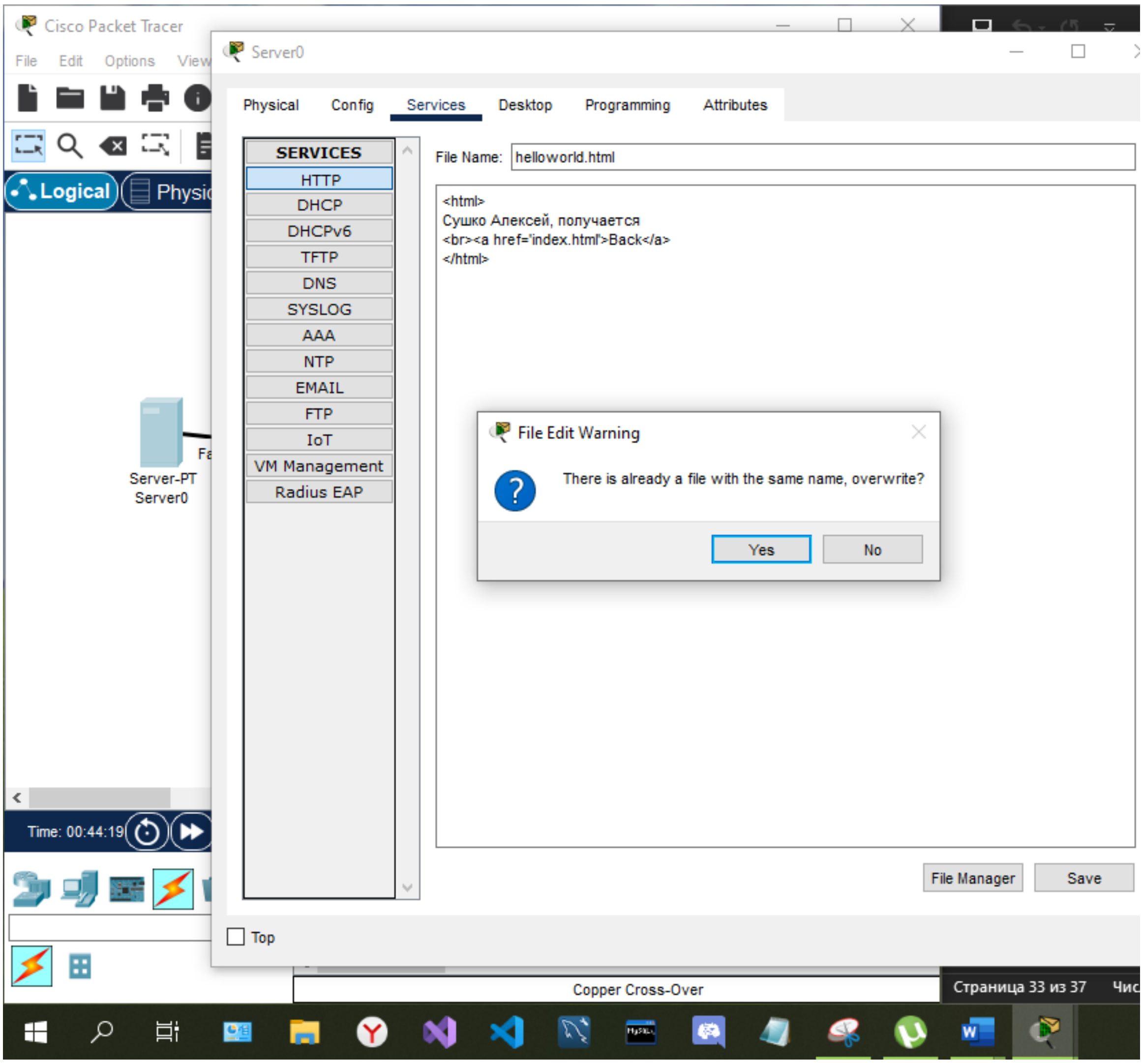


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 417, y: 253
Root
21:32:00
1841
Router0
Fa0/1
Fa0/0
Fa0/1
Server-PT
Server0
Fa0
Fa0/2
Fa0/5
Fa0/4
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.000 --
0.001 PC0
Reset Simulation
Constant Delay
Captured to:
0.001 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 00:41:48.477
PLAY CONTROLS:
Event List
Realtime
Simulation
Copper Cross-Over

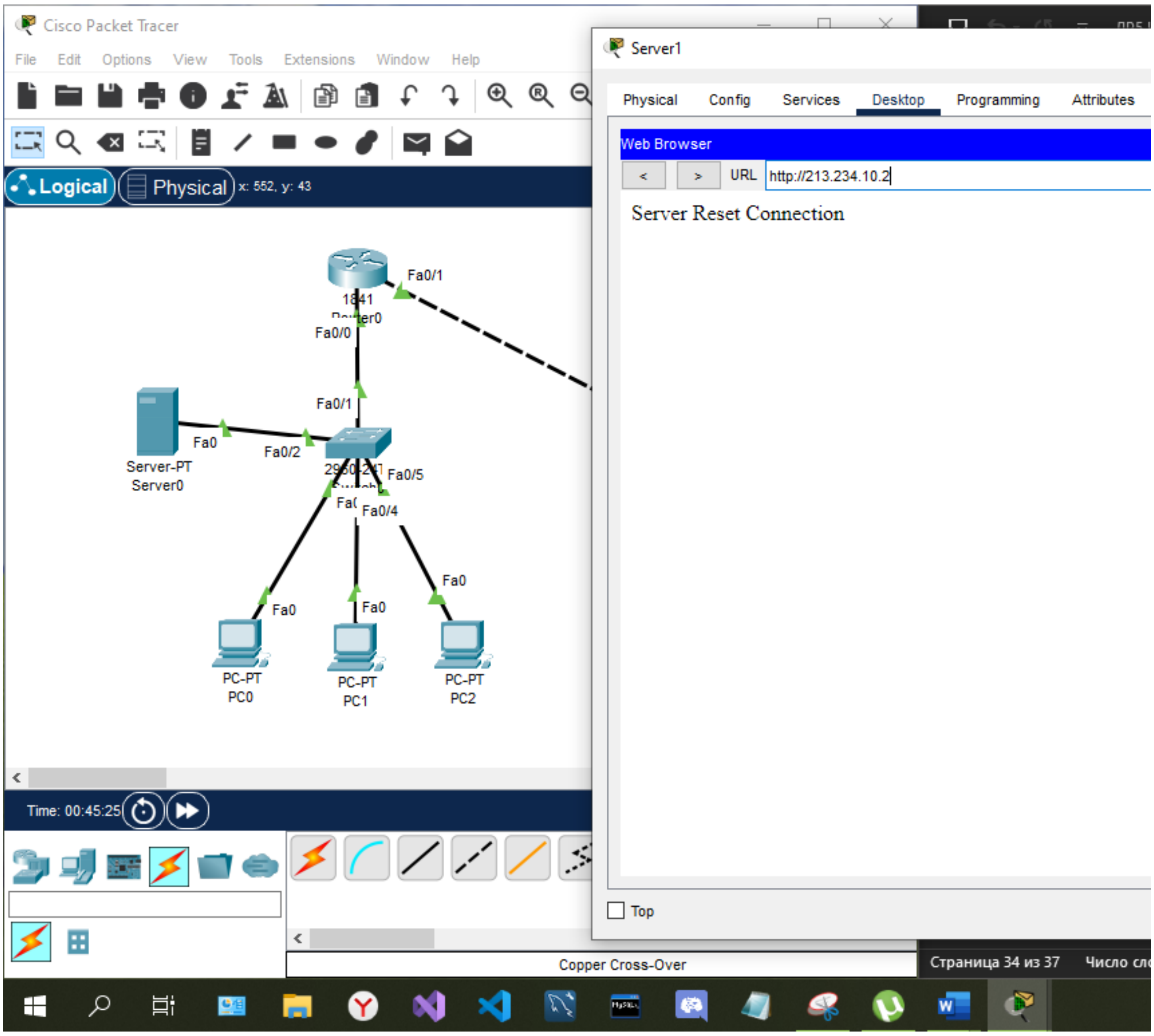


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 538, y: 412
Root
22:02:30
Fa0/1
Fa0
Fa0/0
Fa0/1
1841
Router1
Fa0/5
Fa0/4
Fa0
Fa0
PC-PT
PC2
Simulation Panel
Event List
Vis. Time(sec) Last Device
0.002 Switch0
0.003 Router0
0.004 Router1
Reset Simulation
Constant Delay
Captured to:
0.004 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 00:42:12.995
PLAY CONTROLS:
Event List
Realtime
Simulation
Copper Cross-Over

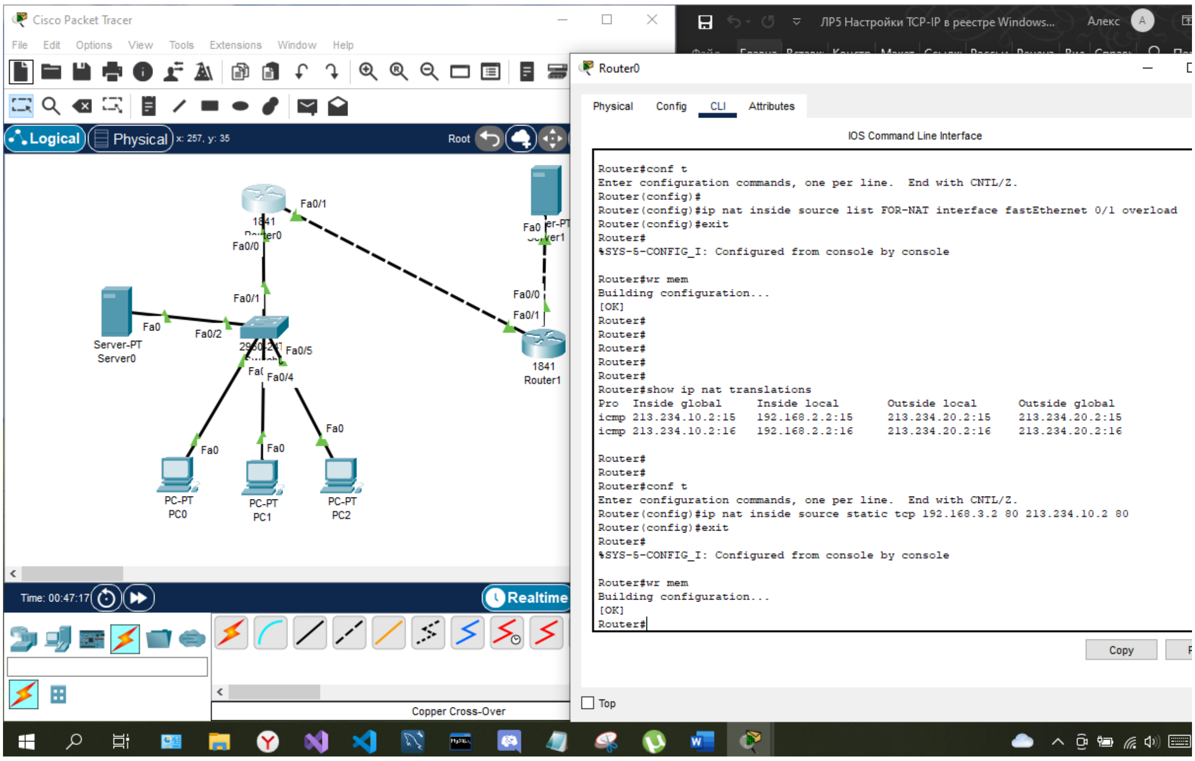

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
Server0
Physical Config Services Desktop Programming Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
HTTP
HTTPS
On
Off
File Manager
File Name
Edit
Delete
1 copyrights.html (edit) (delete)
2 cscoptlogo177x111.jpg (delete)
3 helloworld.html (edit) (delete)
4 image.html (edit) (delete)
5 index.html (edit) (delete)
New File
Import
Top
Server-PT
Server0
PC-PT
PC0
PC1
PC2
Time: 00:42:45
Copper Cross-Over
Страница 32 из 37
Число слов: 1427


Cisco Packet Tracer
File Edit Options View
Logical
Server0
Physical Config Services Desktop Programming Attributes
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
File Name: helloworld.html
<html>
Сушко Алексей, получается

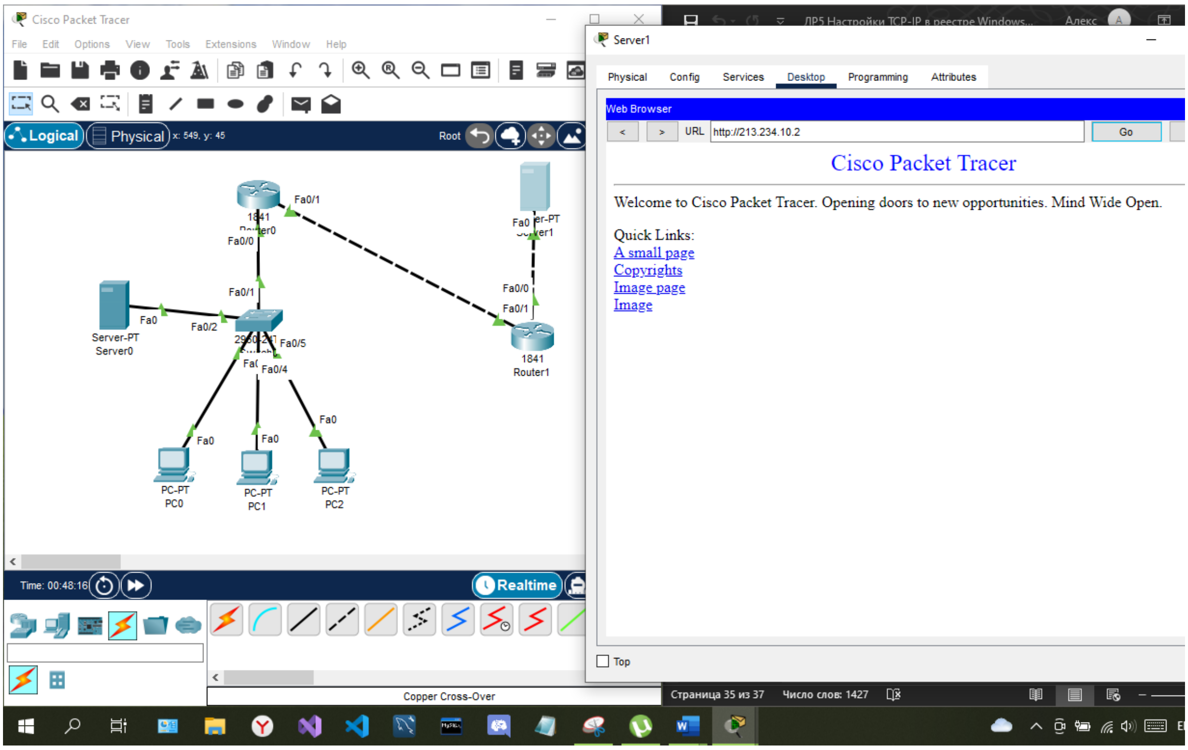
Back
</html>
File Edit Warning
There is already a file with the same name, overwrite?
Yes
No
File Manager
Save
Top
Server-PT
Server0
Time: 00:44:19
Copper Cross-Over
Страница 33 из 37

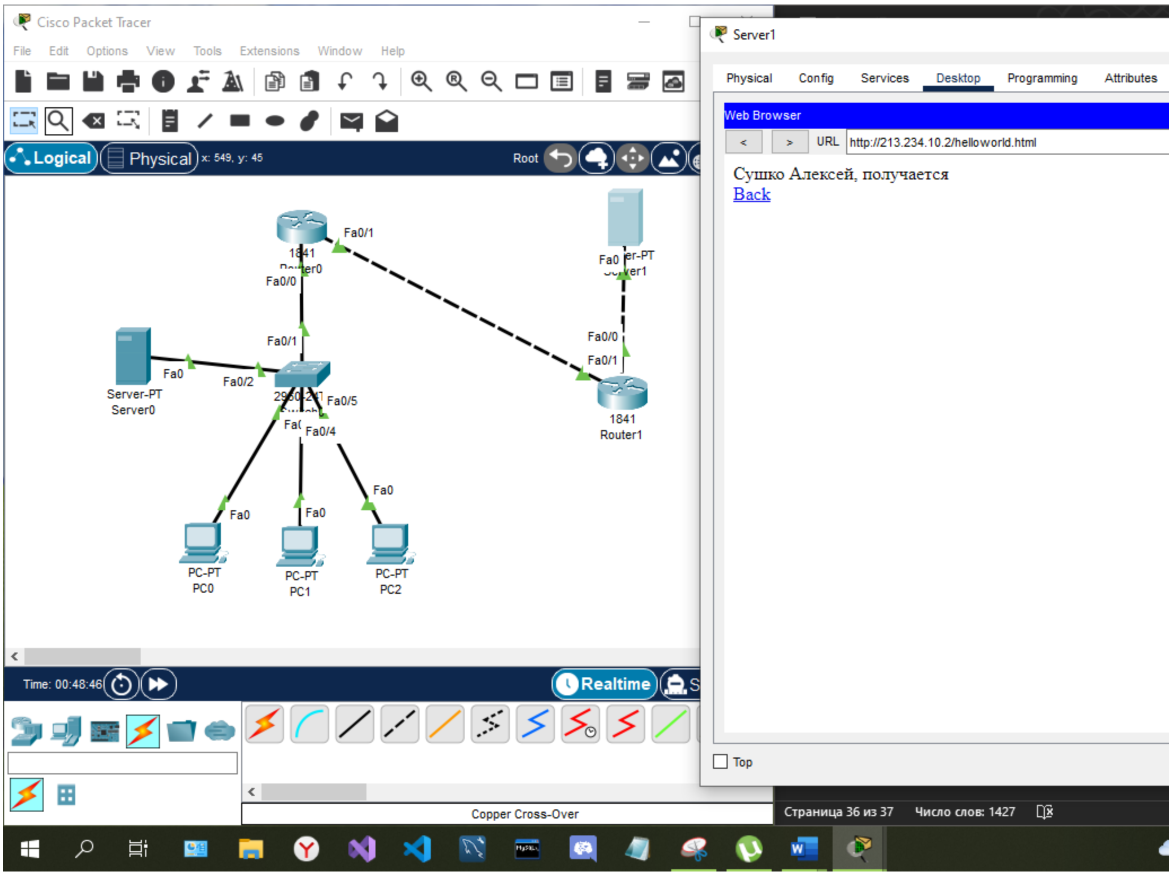


Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 552, y: 43
Fa0/1
1841
Router0
Fa0/0
Fa0/1
Fa0
Fa0/2
Server-PT
Server0
Fa0/5
Fa0/4
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Time: 00:45:25
Copper Cross-Over
Server1
Physical
Config
Services
Desktop
Programming
Attributes
Web Browser
URL
http://213.234.10.2
Server Reset Connection
Top
Страница 34 из 37



Cisco Packet Tracer
File
Edit
Options
View
Tools
Extensions
Window
Help
Logical
Physical
x: 257, y: 35
Root
Fa0/1
1841
Router0
Fa0/0
Fa0/1
Fa0
Fa0/2
Server-PT
Server0
Fa0/5
Fa0/4
Fa0
Fa0
Fa0
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
Fa0/0
Fa0/1
1841
Router1
Time: 00:47:17
Realtime
Copper Cross-Over
ЛР5 Настройки TCP-IP в реестре Windows...
Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#ip nat inside source list FOR-NAT interface fastEthernet 0/1 overload
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Router#
Router#
Router#
Router#
Router#show ip nat translations
Pro Inside global   Inside local   Outside local   Outside global
icmp 213.234.10.2:15   192.168.2.2:15   213.234.20.2:15   213.234.20.2:15
icmp 213.234.10.2:16   192.168.2.2:16   213.234.20.2:16   213.234.20.2:16
Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip nat inside source static tcp 192.168.3.2 80 213.234.10.2 80
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#wr mem
Building configuration...
[OK]
Router#
Copy
Top





Контрольные вопросы

1. Что такое реестр и для чего он нужен?

Реестр Windows – древовидная база данных, которая содержит в себе информацию обо всех параметрах, которые требуются для правильной и бесперебойной работы операционной системы.

2. За что отвечает раздел – HKEY_LOCAL_MACHINE (HKLM)?

   HKEY_LOCAL_MACHINE (HKLM) – ветвь для хранения сведений о загрузке ОС Windows, информации о драйверах устройств и аппаратном обеспечении компьютера.

3. Что влияет на скорость работы Windows?

   Скорость работы системы и ее стабильность напрямую зависит от состояния реестра.

4. Дайте определение термину «свопирование».

   Свопирование - часть информации из ОЗУ записывается на жесткий диск.

Вывод: я научился задавать настройки TCP/IP в реестре Windows.